

# Manual

## ERP Interface with Additional Data EIS-ERP 8.2

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# 1 MF Schnittstelle MES <=> ERP

## Use options

The EIS-ERP interface is used to connect MES systems to superior ERP systems.

This covers the following general requirements:

- Default interface for the adoption of order sequencing data from the ERP system and confirmation/upload of recorded postings in the production.
- Rollout-capable solution.
- Close interfacing of HYDRA and PPS in order to account for the increasing sensitivity of the integrated solution.
- Maintenance options and administration of the interface.

This document describes the principles of the data exchange on the logical level and the system components that are required to this end. In addition, it offers a detailed overview of the interface's structure.

## Implementation notes

Use the interface if

- you wish to automate the exchange of order and operation data between ERP and MES;
- you wish to automate the exchange of confirmations recorded in the MES to the ERP.

## Integration

The interface constitutes the basis for the orders and operations administered in BDE.

And the data recorded in the BDE/shop floor data collection module in turn, constitute the basis for the confirmation/upload to the ERP system.

## Scope of functions

- Transfer of order data
  - Interface enabling the communication between HYDRA and ERP/ PPS systems or external control stations/shop floor scheduling systems.

- Used to interface HYDRA with SAP R/3 PP, Baan IV, Baan ERP, ProAlpha, M-Suite (Soft-M), Brain XPPS, Brain AS, Oracle, PSI PENTA, Navision, ABAS ERP, SYS APS, Infor, IFR ERP, MS-Business Solutions, Axapta, MAPICS, MAS90, Orgaplan, FOSS (ORDAT), I/OPEN, FORS (Atos), DIALOG TOTAL, DTM / KIAS, Siline, IFAI / DIAFERT, MAN/X, FRIDA Command, IFAX Open, Compu Orga, JDE5 (J.D. Edwards), COMET, Bäurer b2 and much more
  - Adoption of released orders and operations
- Transfer of operation confirmations/uploads
  - Confirmation/upload of the collected actual times and quantities referred to transactions/ time tickets
- Transfer of additional data
  - Extension of additional information from ERP
  - Adoption of component data, production resources and tools and document links to the operation
  - Adoption of long texts about the order header and operation
  - Adoption of customer-specific fields referring to the order header and to the operation
- Monitoring
  - Comfortable monitoring and logging functions for the data transfer between HYDRA and the external systems

## 2 Setup of Data Record Structure

The data are transferred in the following structure. Within this structure the value of the SEGNAM field precisely defines the set-up of the user data structure in the SDATA field.

| Field name | Type | Length | Designation           | Data field and meaning                                                                                                                                                        |
|------------|------|--------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEGNAM*    | Char | 30     | Segment name          | This field is occupied by the writing system with the respective segment name. This precisely defines the set-up of the data record (SDATA field).<br>Example: HY72_AU_HD_001 |
| MANDT*     | Char | 3      | Client                | Reserved; fixed: '000'                                                                                                                                                        |
| DOCNUM*    | Char | 16     | IDOC number           | Serial number for the IDOCs<br>Reserved: fixed '0000000000000000'                                                                                                             |
| SEGNUM*    | Char | 6      | Segment number        | Reserved: fixed '000000'                                                                                                                                                      |
| PSEGNUM    | Char | 6      | Parent segment number | Reserved; fixed: '000000'                                                                                                                                                     |
| HLEVEL     | Char | 2      | Hierarchy level       | Reserved; fixed: '00'                                                                                                                                                         |
| SDATA      | Char | 1000   | User data             | This field contains the user data. The structure of this field is defined by the SEGNAM field.                                                                                |

### 3 Data type definitions

|                                   | →                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Type                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| <b>CHAR x</b>                     | <p>The information is left-aligned for the data type CHAR; unnecessary places are filled with blanks (blanks - (U+0020)).</p> <p>If a field is not used, fill it in full length with blanks.</p> <p>Example: "ABCD "</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
| <b>NUM x</b>                      | <p>Numeric field of the length x without sign. The NUMC data type only supports digits (ASCII characters 30 hex to 39 hex and/or U+0030 – U+0039). Numbers are right-aligned; unnecessary places are filled with zeros (U+0030).</p> <p>If a field is not used, fill it in full length with zeros.</p> <p>Example: "00000002"</p>                                                                                                                                                                                                                                                                                                                                                          |  |
| <b>DEC x.y</b><br><b>QUAN x.y</b> | <p>Numeric field of the length x and y decimal places. A data field in HYDRA format is <b>preceded</b> by a sign ("+" or "-") and includes a decimal point. Enter zeros to fill empty places.</p> <p>If a field is not used, fill it in full length with zeros (U+0030) including sign and decimal separator.</p> <p>e.g. DEC 13,3:</p> <ul style="list-style-type: none"> <li>-1234567890,123 → -1234567890.123</li> <li>234567890,3 → +0234567890.300</li> </ul> <p>Note:</p> <p>The field length is indicated WITHOUT algebraic sign and WITHOUT decimal point in the tabular description of the structure. For example: a QUAN 13.3 field results in an external length of CHAR15.</p> |  |
| <b>DATE</b>                       | <p>Dates must be transferred in the HYDRA format <b>MM/DD/YYYY</b>.</p> <p>Populate unused date fields with blanks (U+0020; zero(s) (U+0030) not accepted).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
| <b>TIME</b>                       | <p>Times must be transferred in the HYDRA format <b>seconds after midnight</b> (0 - 86400).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |



For all alphanumeric fields, HYDRA does not support specific special characters. These characters are: "\" (backslash - U+005B), "|" (pipe - U+007C), „ “ “ (double quote - U+0022), and „ ’ “ (single quote - U+0027). You cannot enter these characters using the shop floor terminals;

the terminals and the MOC do not support these characters.



The character " ; " (semicolon - U+003B) is used as separator for data collection. You must not use this character in key fields (e.g. order, batch number, personnel number, etc.).



The character " % " (percent - U+0025) is used as placeholder/wildcard character for database queries. For this reason, you should avoid using this character as it might falsify results.



In general, you must not use special characters ranging from U+0000 to U+001F. Exception: U+000A and U+000D as end-of-line characters.



The file must not include [Byte Order Mark \(BOM\)](#).

In general, HYDRA always expects a contiguous data structure. Consequently, you have to populate unused data fields with such default values that comply with the applicable conventions. This also applies to fields that are not required at the end of a data structure. The following definitions apply if you use the file port:

Each data record included in the file has to be completed by 'CR' (U+000D) und 'LF' (U+000A) for Windows and 'LF' (U+000A) for Unix.

HYDRA expects the file to be in the UTF-8 format and HYDRA also uses this format for uploads. On request, you can also transfer files in the file format that was used until MW 2.x.



## 4 Key Fields / Supported Characters

### General notes

In all alphanumeric fields, HYDRA does not support specific special characters. The following characters are not supported: "%", "ß", "\*", "\", "/", "|", "\_", "?". Reason: You cannot enter these characters on the shop floor terminals or the clients do not support these characters.

You must not use the characters ";" (semicolon), "," (comma) and "'" (apostrophe) because they are often interpreted as comment characters or separators and can lead to unwanted results.

### Workplace/machine numbers (resources of type "MNR")

Workplace/machine numbers and numbers of capacity/machine groups are interpreted as alphanumeric values. Alphanumeric field with a maximum length of 8, left-aligned.

When you create or copy a workplace, the system checks if the characters used are allowed. The following characters are allowed:

- Numbers "0" to "9" ([US-ASCII 30<sub>hex</sub> - 39<sub>hex</sub>](#))
- Letters "A" - "Z" (upper case letters - [US-ASCII 41<sub>hex</sub> - 5A<sub>hex</sub>](#))
- "-" ([US-ASCII 2D<sub>hex</sub>](#))

Lower case letters are automatically converted to upper case letters when a new workplace is created.

You must not use blanks. If required, you must prefix the numbers by leading zeros ("0").

The entry "SYSTEM" as workplace/machine number is reserved for HYDRA and may not be used.

It is possible to overwrite the valid characters for the workplace/machine numbers in the INI configuration. To this end, you must specify the valid characters as a regular expression (in brackets).

| Field   | Value                                     |
|---------|-------------------------------------------|
| Name    | INPUT                                     |
| Section | PATTERN                                   |
| Key     | MNR                                       |
| Value   | ^(?!SYSTEM) ([A-Z0-9(){}~^#+!\$. _%-]+)\$ |
| Active  | <input checked="" type="checkbox"/>       |

Minimum requirement: b\_mnr.dll version 8.1.1.102

## Resource numbers (resources of type <> "MNR")

Resource numbers are interpreted as alphanumeric values. Alphanumeric field with a maximum length of 20; left-aligned.

When you create or copy a resource, the system checks if the characters used are allowed. The following characters are allowed:

- Numbers "0" to "9" ([US-ASCII 30<sub>hex</sub> - 39<sub>hex</sub>](#))
- Letters "A" - "Z" (upper case letters - [US-ASCII 41<sub>hex</sub> - 5A<sub>hex</sub>](#))
- Umlauts "Ä", "Ö", "Ü" ([Extended ASCII C4<sub>hex</sub>, D6<sub>hex</sub>, DC<sub>hex</sub>](#))
- "-" ([US-ASCII 2D<sub>hex</sub>](#))

You may not use umlauts or special characters (e.g. "%", "ß", "\*", "\", "/", "|", "\_", "?") because you cannot enter these characters on the shop floor terminals or because the clients do not support these characters.

You must not use blanks. If required, you must prefix the numbers by leading zeros ("0").

Lower case letters are automatically converted to upper case letters when a new resource is created.

It is possible to overwrite the valid characters for the resource numbers in the INI configuration. To this end, you must specify the valid characters as a regular expression (in brackets).

| Field   | Value                               |
|---------|-------------------------------------|
| Name    | INPUT                               |
| Section | PATTERN                             |
| Key     | RES                                 |
| Value   | ^([0-9A-ZÄÖÜ(){}~^#+!\$. _%-]+)\$   |
| Active  | <input checked="" type="checkbox"/> |

Minimum requirement: b\_res.dll version 8.1.1.117

## HYDRA order number

There are some differences with respect to the order number in the HYDRA data model and the interface.

### Order number

The order number (field AUNR) contains the actual order number as it is known in the ERP system and transferred to HYDRA. The order number is specified in the HYDRA basic settings; by default, this number has a length of 8 characters.

### Operation number

The operation number (field AGNR) clearly identifies a defined process step of an order. The operation number is specified in the HYDRA basic settings; by default, this number has a length of 4 characters.

**MES order number**

The MES order number (field ANR) combines the order and the operation number and sometimes also the sequence number from the ERP system (if licensed). Its length therefore results from the total of the separate number lengths.

The total length must not exceed 25 digits. If DOS terminals are used, the total length must not exceed 16 digits.

**Note the following for the order or operation number:**

- Preferably only use the numbers "0" to "9" ([US-ASCII 30<sub>hex</sub>](#) - [39<sub>hex</sub>](#)).
- If you use letters, only the characters "A" - "Z" (upper case - [US-ASCII 41<sub>hex</sub>](#) - [5A<sub>hex</sub>](#)) and "-" ([US-ASCII 2D<sub>hex</sub>](#)) are allowed. Do not use lower case letters.
- You may not use blanks in the numbers. The order or operation numbers must have the specified number of digits with the characters "0" to "9" or "A" to "Z". If required, you must prefix the numbers by leading zeros ("0").
- HYDRA does not support any umlauts, blank or special characters (see section *General notes*) for the order or operation number because you cannot enter these characters on the shop floor terminals or because the clients do not support these characters.

## 5 Order Data ERP --> HYDRA

### Data transfer process

Data of the ERP system is provided in a file that must be available in a defined directory of the HYDRA server. The configuration is made in the application ["Logical Systems"](#). You can combine the data transferred from the ERP system to HYDRA in one file to create/change data or to delete data. HYDRA uses the segment name of a data record (one line in the file) to identify if the respective data record must be added/updated (suffix "\_A" for segment name) or if it must be deleted (suffix "\_D" for segment name).

HYDRA processes the contents of the file sequentially top down. The higher a data record is placed in the file, the earlier it will be processed in HYDRA. This ensures that the file can be created in the chronologically correct order in the ERP system.



HYDRA 8 still supports but no longer improves the HYDRA 7 option ("compatibility mode") of transferring data to be created/changed and data to be deleted in separate files using different file extensions.

### Create/change data

The interface transferring production orders bundles all data relevant to HYDRA in an IDOC and sends this data to HYDRA using the message function (= file extension) "DAT".

Make sure to transfer the data in a specific order, when transferring order data from the ERP system to HYDRA for the first time. This means that the order data described in the following sections must be transferred to HYDRA in the specified sequence/hierarchy. Within a transfer level, you are free to specify any sequence, e.g. you can transfer operation 0200 before operation 0100.

By default, you are not allowed to change operations with the status "running", "finished" or "deleted".

In general, HYDRA always expects a consistent data structure. This means: If a data field is not used, the predefined default value is assigned.



An operation in status "finished" is set to status "interrupted" when the operation is transferred again (update), if the following conditions are fulfilled:

- The operation has been deleted via ERP interface.
- The operation status set at the time of "delete": For this status, the action "delete" was configured in the order status configuration with "E" or "X".
- The operation is still in the online data set when transferred again (update).

If one of the conditions is not fulfilled, then an operation in status "finished" is not set to status "interrupted" when transferred again (update). The above also applies if the operation has been logged off via posting functionality (e.g. operation logoff).

## Delete data

If entire orders are deleted in the ERP system, an IDOC is generated and sent to HYDRA using the message function (= file extension) "DAT". The data to be deleted must be transferred in the following order:

- Long texts of the operation
- Documents
- Production resources and tools
- Components
- Operation

When the last operation of an order is deleted via the interface , the order header is also deleted automatically<sup>1</sup>.

If the ERP system transfers the deletion of an order header, all operations and the included data are deleted.

For deletion, it is sufficient to just indicate the respective key fields in the segment.

You cannot delete operations assigned to the operation statuses *Running*, *Interrupted*, *Finished* or *Deleted*.

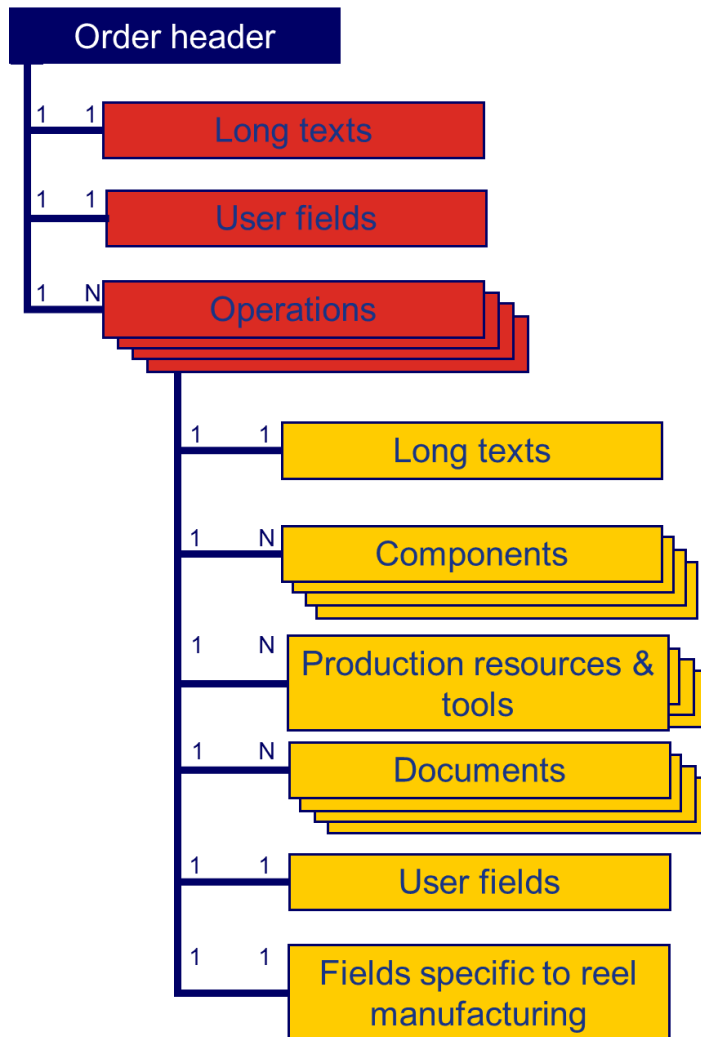
You can neither delete operations that are split or part of a merged operation.

## Data record definition

Order-related data is transferred for each production order in a multi-level IDOC. This IDOC structure is as follows:

---

<sup>1</sup> If the last operation of an order is deleted via the MOC, the order header is not automatically deleted.



The structures that belong to the individual segments are presented in the following. The individual columns have the following meaning:

| Column           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Field            | Name of the field                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| V (usage)        | <p>S Key field clearly identifying the data record. (Further key fields might be required). The field must be populated.</p> <p>M Mandatory field which must be populated with a valid value.</p> <p>ML Mandatory field if the HYDRA Shop Floor Scheduling is used (HLS).</p> <p>MM Mandatory field if the HYDRA Material and Production Logistics (MPL and/or MPL/RF) is used.</p> <p>K Field may stay empty (optional field).</p> <p>SA Mandatory field if the Arburg control system (SCS-ALS) is in use; otherwise optional field.</p> |
| T(type)          | <a href="#">Data type</a> of the field                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| L(ength)         | Field length<br>For fields of data type DEC: Total number of digits without decimal separator and algebraic sign.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| D(ecimal places) | For fields of data type DEC: Number of decimal places; otherwise: not relevant                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Description      | Field description and/or comments on the field.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Segments to create/modify data

The following specifications arise for the IDOC:

**Message type /  
file name:**

HY72PPS

**Message  
file extensions:**

**functions/** DAT<sup>2</sup>

**Segments:**

```

HY72_AU_HD_001_A (order header)
├ HY72_AU_INFO_AI_001_A (long texts)
├ HY72_AU_USRFLD_001_A (user fields)
├ HY72_AG_HD_001_A (operation data – part 1)
│   ├── HY72_AG_HD_002_A (operation data – part 2)
│   ├── HY72_AG_KOMPL_001_A (component list)
│   │   └ HY72_AG_KOMPL_USRFLD_001_A (comp. user fields)
│   ├── HY72_AG_FHM_001_A (PRT / resources)
│   ├── HY72_AG_DOC_001_A (documents)
│   ├── HY72_AG_INFO_AI_001_A (long texts)
│   ├── HY72_AG_USRFLD_001_A (user fields)
│   └ HY72_AG_RF_001_A (MPL for data specific to coil-based
manufacturing)
└ HY72_FERTVAR_001_A (production variants)

```

## Segments to delete data

The following specifications arise for the IDOC:

**Message type /  
file name:**

HY72PPS

**Message  
file extensions:**

**functions/** DAT

**Segments:**

```

HY72_AU_HD_001_D (order header)
├ HY72_AU_INFO_AI_001_D (long texts)
└ HY72_AG_HD_001_D (operation data)
    ├── HY72_AG_KOMPL_001_D (component list)
    ├── HY72_AG_FHM_001_D (PRT / resources)
    ├── HY72_AG_DOC_001_D (documents)
    └ HY72_AG_INFO_AI_001_D (long texts)

```

## Order header

<sup>2</sup> HYDRA inbound processing discards the file extension "DAT" (importing the file and storage in MLE inbound transactions). The MLE inbound transactions display a data record of such a file without message function.

|               |                                      |
|---------------|--------------------------------------|
| Segment name: | HY72_AU_HD_001_A<br>HY72_AU_HD_001_D |
|---------------|--------------------------------------|

The order header includes the header data for the production order.

| Field    | V  | T    | L  | D | Description                                                                                                                                                                                      | Fro<br>m <sup>3</sup> | To <sup>4</sup> |
|----------|----|------|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------|
| AUNR     | S  | CHAR | 40 |   | Order number                                                                                                                                                                                     | 1                     | 40              |
| AUART    | M  | CHAR | 5  |   | Order type<br>HYDRA order type (see note)                                                                                                                                                        | 41                    | 45              |
| ATK      | M  | CHAR | 40 |   | Article;<br>alphabetic characters in CAPITAL LETTERS                                                                                                                                             | 46                    | 85              |
| ATKBEZ   | K  | CHAR | 40 |   | Article designation                                                                                                                                                                              | 86                    | 125             |
| KDBEZ    | K  | CHAR | 40 |   | Customer name                                                                                                                                                                                    | 126                   | 165             |
| KDAUNR   | K  | CHAR | 25 |   | Sales order<br>Note: Using the MOC, you can only edit up to 20 digits/characters.                                                                                                                | 166                   | 190             |
| KDAUPOS  | K  | CHAR | 15 |   | Sales order item                                                                                                                                                                                 | 191                   | 205             |
| EXTPRIO  | K  | CHAR | 1  |   | Priority                                                                                                                                                                                         | 206                   | 206             |
| AUIDX    | K  | DEC  | 4  | 2 | Order index should be 0 (transfer: "+00.00")<br>Please note: Make sure to add a sign and a decimal point to the length specified here. This applies to all following fields of the DEC data type | 207                   | 212             |
| SGE:B    | M  | CHAR | 3  |   | Base quantity unit = quantity unit of the order header<br>This quantity unit will be used for operations.                                                                                        | 213                   | 215             |
| SGR:GUTB | M  | DEC  | 13 | 3 | Target quantity (base quantity unit)                                                                                                                                                             | 216                   | 230             |
| SGR:AUSB | K  | DEC  | 13 | 3 | Target scrap quantity (base quantity unit)                                                                                                                                                       | 231                   | 245             |
| MATTYP   | K  | CHAR | 10 |   | Material type of the article<br>If HYDRA MPL is used, enter a valid material type configured in HYDRA.<br>otherwise: "SYSTEM"                                                                    | 246                   | 255             |
| FILLER   | K  | CHAR | 10 |   | reserved; should be empty                                                                                                                                                                        | 256                   | 265             |
| CNR      | K  | CHAR | 20 |   | Batch number                                                                                                                                                                                     | 266                   | 285             |
| PCNR     | K  | CHAR | 20 |   | Inspection order/inspection batch number                                                                                                                                                         | 286                   | 305             |
| PPKTTYP  | K  | CHAR | 1  |   | Sample type                                                                                                                                                                                      | 306                   | 306             |
| DATFB    | ML | DATE | 10 |   | Earliest start (date)                                                                                                                                                                            | 307                   | 316             |
| ZEIFB    | ML | TIME | 5  |   | Earliest start (time)                                                                                                                                                                            | 317                   | 321             |
| DATSE    | ML | DATE | 10 |   | Latest end (date)                                                                                                                                                                                | 322                   | 331             |
| ZEISE    | ML | TIME | 5  |   | Latest end (time)                                                                                                                                                                                | 332                   | 336             |

<sup>3</sup> The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA application developers. It does not include the number of characters of the file itself but only of the field SDATA.

<sup>4</sup> See footnote of column "From"



| Field          | V | T    | L  | D | Description                                                                                                                          |                                                                                                                                                                                                                                                                              | Fro<br>m³ | To⁴ |
|----------------|---|------|----|---|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----|
| DATTERM        | K | DATE | 10 |   | Scheduled start (date)                                                                                                               | If scheduling is run outside HYDRA, the scheduled dates of the order (header) should be transferred.<br><br>Note: For the processing, these dates are only used for information purposes in HYDRA.<br><br>If scheduling is performed in HYDRA, these fields are overwritten. | 337       | 346 |
| ZEITERM        | K | TIME | 5  |   | Scheduled start (time)                                                                                                               |                                                                                                                                                                                                                                                                              | 347       | 351 |
| DATERME        | K | DATE | 10 |   | Scheduled end (date)                                                                                                                 |                                                                                                                                                                                                                                                                              | 352       | 361 |
| ZEITERME       | K | TIME | 5  |   | Scheduled end (time)                                                                                                                 |                                                                                                                                                                                                                                                                              | 362       | 366 |
| TERMART        | K | CHAR | 1  |   | Scheduling type; mandatory field if scheduling is to be made in HYDRA.<br>(V=forward scheduling, R=backward scheduling)              |                                                                                                                                                                                                                                                                              | 367       | 367 |
| REDSTRAT       | K | CHAR | 2  |   | Reduction strategy, according to configuration.                                                                                      |                                                                                                                                                                                                                                                                              | 368       | 369 |
| AUGRP          | K | CHAR | 4  |   | Order group                                                                                                                          |                                                                                                                                                                                                                                                                              | 370       | 373 |
| DISP           | K | CHAR | 10 |   | MRP controller                                                                                                                       |                                                                                                                                                                                                                                                                              | 374       | 383 |
| PRJNR          | K | CHAR | 25 |   | Project number                                                                                                                       |                                                                                                                                                                                                                                                                              | 384       | 408 |
| PLANAUNR       | K | CHAR | 25 |   | Planned order                                                                                                                        |                                                                                                                                                                                                                                                                              | 409       | 433 |
| KTR            | K | CHAR | 25 |   | Cost object                                                                                                                          |                                                                                                                                                                                                                                                                              | 434       | 458 |
| APNR           | K | CHAR | 40 |   | Work plan                                                                                                                            |                                                                                                                                                                                                                                                                              | 459       | 498 |
| APVER          | K | CHAR | 12 |   | Work plan version                                                                                                                    |                                                                                                                                                                                                                                                                              | 499       | 510 |
| SLVER          | K | CHAR | 12 |   | BOM version                                                                                                                          |                                                                                                                                                                                                                                                                              | 511       | 522 |
| KLKK:MNR       | K | DEC  | 13 | 3 | Calc. costs - machines                                                                                                               |                                                                                                                                                                                                                                                                              | 523       | 537 |
| KLKK:L         | K | DEC  | 13 | 3 | Calc. costs - payroll                                                                                                                |                                                                                                                                                                                                                                                                              | 538       | 552 |
| KLKK:MAT       | K | DEC  | 13 | 3 | Calc. costs - material                                                                                                               |                                                                                                                                                                                                                                                                              | 553       | 567 |
| KLKK:SONST     | K | DEC  | 13 | 3 | Calc. costs - other                                                                                                                  |                                                                                                                                                                                                                                                                              | 568       | 582 |
| MATWERT:GUT    | K | DEC  | 13 | 3 | Material value                                                                                                                       |                                                                                                                                                                                                                                                                              | 583       | 597 |
| MATWERT:AUS    | K | DEC  | 13 | 3 | Scrap value                                                                                                                          |                                                                                                                                                                                                                                                                              | 598       | 612 |
| ANR.KBN:LBEZID | K | CHAR | 15 |   | eKANBAN control cycle<br>This field is available as of the MLE variant HY72PPS_018                                                   |                                                                                                                                                                                                                                                                              | 613       | 627 |
| ATKIDX         | K | CHAR | 50 |   | Order information → Order header (AK) → Drawing issue number<br><br>This field is available as of BDE82 and MLE version HY72PPS_023. |                                                                                                                                                                                                                                                                              | 628       | 677 |



In HYDRA the order type controls the global behavior of an order. The entered order type must exist in HYDRA and be entirely configured.

## Long texts of the order header

|               |                                                |
|---------------|------------------------------------------------|
| Segment name: | HY72_AU_INFO_AI_001_A<br>HY72_AU_INFO_AI_001_D |
|---------------|------------------------------------------------|

Use the following structure to transfer text fields as additional information to the order(header) to HYDRA. The MOC will then show this additional information. Each data record contains one page of a maximum of 10 lines and 80 characters of text information each.

The long text of the order header is shown in the MOC application "edit orders".

| Field    | V | T    | L  | D | Description                                                                       |
|----------|---|------|----|---|-----------------------------------------------------------------------------------|
| KEY      | S | CHAR | 40 |   | Order number (AUNR)                                                               |
| TYP      | S | CHAR | 2  |   | Record type; by default: "AI"                                                     |
| SUBKEY:1 | S | NUM  | 8  |   | Reserved; assign "00000000" by default.                                           |
| SUBKEY:2 | M | NUM  | 8  |   | Consecutive numbering within the key starting at "00000001".                      |
| INFO:BEZ | K | CHAR | 20 |   | Short text; if empty, the first 20 digits/characters of Info text 1 will be used. |
| INFO:1   | K | CHAR | 80 |   | Info text 1                                                                       |
| INFO:2   | K | CHAR | 80 |   | Info text 2                                                                       |
| INFO:3   | K | CHAR | 80 |   | Info text 3                                                                       |
| INFO:4   | K | CHAR | 80 |   | Info text 4                                                                       |
| INFO:5   | K | CHAR | 80 |   | Info text 5                                                                       |
| INFO:6   | K | CHAR | 80 |   | Info text 6                                                                       |
| INFO:7   | K | CHAR | 80 |   | Info text 7                                                                       |
| INFO:8   | K | CHAR | 80 |   | Info text 8                                                                       |
| INFO:9   | K | CHAR | 80 |   | Info text 9                                                                       |
| INFO:10  | K | CHAR | 80 |   | Info text 10                                                                      |

## User fields of the order header

|               |                      |
|---------------|----------------------|
| Segment name: | HY72_AU_USRFLD_001_A |
|---------------|----------------------|

Apart from the fields that are available by default, you can use user fields to store further customer-specific information to HYDRA. Use this segment to transfer this data from the ERP system to HYDRA and to store this data in the order(header).

The so-called user field key specifies the available user fields and their meaning. Each user field key describes a combination of user fields. The document [Configuration\\_Userfields.pdf](#) describes how to configure the user field key.



To allow for a consistent data exchange between the ERP system and HYDRA, the customer must ensure that the user field keys are identical in both systems.

| Field  | V | T    | L  | D | Description                                                          | From <sup>5</sup> | To <sup>6</sup> |
|--------|---|------|----|---|----------------------------------------------------------------------|-------------------|-----------------|
| AUNR   | S | CHAR | 40 |   | Order number                                                         | 1                 | 40              |
| USRFLD | S | CHAR | 8  |   | User field key. The user field key must be configured in the system. | 41                | 48              |
| FU:1   | K | DATE | 10 |   | User field 1                                                         | 49                | 58              |
| FU:2   | K | DATE | 10 |   | User field 2                                                         | 59                | 68              |
| FU:3   | K | DATE | 10 |   | User field 3                                                         | 69                | 78              |
| FU:4   | K | DATE | 10 |   | User field 4                                                         | 79                | 88              |
| FU:5   | K | DATE | 10 |   | User field 5                                                         | 89                | 98              |
| FU:6   | K | DATE | 10 |   | User field 6                                                         | 99                | 108             |
| FU:7   | K | NUM  | 8  |   | User field 7                                                         | 109               | 116             |
| FU:8   | K | NUM  | 8  |   | User field 8                                                         | 117               | 124             |
| FU:9   | K | NUM  | 8  |   | User field 9                                                         | 125               | 132             |
| FU:10  | K | NUM  | 8  |   | User field 10                                                        | 133               | 140             |
| FU:11  | K | NUM  | 8  |   | User field 11                                                        | 141               | 148             |
| FU:12  | K | NUM  | 8  |   | User field 12                                                        | 149               | 156             |
| FU:13  | K | NUM  | 8  |   | User field 13                                                        | 157               | 164             |
| FU:14  | K | NUM  | 8  |   | User field 14                                                        | 165               | 172             |
| FU:15  | K | NUM  | 8  |   | User field 15                                                        | 173               | 180             |
| FU:16  | K | NUM  | 8  |   | User field 16                                                        | 181               | 188             |
| FU:17  | K | NUM  | 8  |   | User field 17                                                        | 189               | 196             |
| FU:18  | K | NUM  | 8  |   | User field 18                                                        | 197               | 204             |
| FU:19  | K | NUM  | 8  |   | User field 19                                                        | 205               | 212             |
| FU:20  | K | NUM  | 8  |   | User field 20                                                        | 213               | 220             |
| FU:21  | K | NUM  | 8  |   | User field 21                                                        | 221               | 228             |
| FU:22  | K | NUM  | 8  |   | User field 22                                                        | 229               | 236             |
| FU:23  | K | DEC  | 13 | 3 | User field 23                                                        | 237               | 251             |
| FU:24  | K | DEC  | 13 | 3 | User field 24                                                        | 252               | 266             |

<sup>5</sup> The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA application developers. It does not include the number of characters of the file itself but only of the field SDATA.

<sup>6</sup> See footnote of column "From"

| Field | V | T    | L  | D | Description   | From <sup>5</sup> | To <sup>6</sup> |
|-------|---|------|----|---|---------------|-------------------|-----------------|
| FU:25 | K | DEC  | 13 | 3 | User field 25 | 267               | 281             |
| FU:26 | K | DEC  | 13 | 3 | User field 26 | 282               | 296             |
| FU:27 | K | DEC  | 13 | 3 | User field 27 | 297               | 311             |
| FU:28 | K | DEC  | 13 | 3 | User field 28 | 312               | 326             |
| FU:29 | K | CHAR | 1  |   | User field 29 | 327               | 327             |
| FU:30 | K | CHAR | 1  |   | User field 30 | 328               | 328             |
| FU:31 | K | CHAR | 1  |   | User field 31 | 329               | 329             |
| FU:32 | K | CHAR | 1  |   | User field 32 | 330               | 330             |
| FU:33 | K | CHAR | 1  |   | User field 33 | 331               | 331             |
| FU:34 | K | CHAR | 1  |   | User field 34 | 332               | 332             |
| FU:35 | K | CHAR | 1  |   | User field 35 | 333               | 333             |
| FU:36 | K | CHAR | 1  |   | User field 36 | 334               | 334             |
| FU:37 | K | CHAR | 1  |   | User field 37 | 335               | 335             |
| FU:38 | K | CHAR | 1  |   | User field 38 | 336               | 336             |
| FU:39 | K | CHAR | 1  |   | User field 39 | 337               | 337             |
| FU:40 | K | CHAR | 1  |   | User field 40 | 338               | 338             |
| FU:41 | K | CHAR | 1  |   | User field 41 | 339               | 339             |
| FU:42 | K | CHAR | 1  |   | User field 42 | 340               | 340             |
| FU:43 | K | CHAR | 1  |   | User field 43 | 341               | 341             |
| FU:44 | K | CHAR | 1  |   | User field 44 | 342               | 342             |
| FU:45 | K | CHAR | 10 |   | User field 45 | 343               | 352             |
| FU:46 | K | CHAR | 10 |   | User field 46 | 353               | 362             |
| FU:47 | K | CHAR | 10 |   | User field 47 | 363               | 372             |
| FU:48 | K | CHAR | 10 |   | User field 48 | 373               | 382             |
| FU:49 | K | CHAR | 10 |   | User field 49 | 383               | 392             |
| FU:50 | K | CHAR | 10 |   | User field 50 | 393               | 402             |
| FU:51 | K | CHAR | 20 |   | User field 51 | 403               | 422             |
| FU:52 | K | CHAR | 20 |   | User field 52 | 423               | 442             |
| FU:53 | K | CHAR | 20 |   | User field 53 | 443               | 462             |
| FU:54 | K | CHAR | 20 |   | User field 54 | 463               | 482             |
| FU:55 | K | CHAR | 20 |   | User field 55 | 483               | 502             |
| FU:56 | K | CHAR | 20 |   | User field 56 | 503               | 522             |
| FU:57 | K | CHAR | 20 |   | User field 57 | 523               | 542             |
| FU:58 | K | CHAR | 20 |   | User field 58 | 543               | 562             |
| FU:59 | K | CHAR | 20 |   | User field 59 | 563               | 582             |
| FU:60 | K | CHAR | 20 |   | User field 60 | 583               | 602             |

| Field | V | T    | L  | D | Description   | From <sup>5</sup> | To <sup>6</sup> |
|-------|---|------|----|---|---------------|-------------------|-----------------|
| FU:61 | K | CHAR | 20 |   | User field 61 | 603               | 622             |
| FU:62 | K | CHAR | 20 |   | User field 62 | 623               | 642             |
| FU:63 | K | CHAR | 20 |   | User field 63 | 643               | 662             |
| FU:64 | K | CHAR | 20 |   | User field 64 | 663               | 682             |
| FU:65 | K | CHAR | 40 |   | User field 65 | 683               | 722             |
| FU:66 | K | CHAR | 40 |   | User field 66 | 723               | 762             |

#### Note

This data structure is only available with the respective license.

## Order sequences

|               |                                      |
|---------------|--------------------------------------|
| Segment name: | HY72_AFOLG_001_A<br>HY72_AFOLG_001_D |
|---------------|--------------------------------------|

Only use this segment to transfer data if sequences are in use. This specification is made as part of the customer project.

| Field   | V | T    | L  | D | Description                                                                                                                                                                                                                                             |
|---------|---|------|----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AUNR    | S | CHAR | 40 |   | Order number                                                                                                                                                                                                                                            |
| AFOLG   | S | CHAR | 6  |   | Number of the sequence<br>Number of the sequence. Length as configured in the HYDRA basic settings, left-aligned including leading zeros.<br>Sequence type "S": must be 0<br>Sequence type "P", "A": sequence number                                    |
| FOLGART | S | CHAR | 1  |   | Sequence type:<br>S = master sequence<br>P = parallel sequence<br>A = alternative sequence                                                                                                                                                              |
| AKTIV   | M | CHAR | 1  |   | Active; with alternative sequences J/N; otherwise always J                                                                                                                                                                                              |
| BEZK    | K | CHAR | 40 |   | Short text/description of the sequence: comment field                                                                                                                                                                                                   |
| VER     | K | CHAR | 12 |   | Version; comment field; not processed in HYDRA                                                                                                                                                                                                          |
| BZGFOLG | M | CHAR | 10 |   | Reference sequence.<br>Number of the reference sequence this sequence refers to. Length as configured in the HYDRA basic settings, left-aligned including leading zeros.<br>Sequence type "S": leave empty<br>Sequence type "A", "P": must always be 0. |
| ANRA    | M | CHAR | 40 |   | Branch operation of the reference sequence; combined HYDRA order number<br>Sequence type "S": leave empty                                                                                                                                               |
| ANRR    | M | CHAR | 40 |   | Return OP of the reference sequence; combined HYDRA order number<br>Sequence type "S": leave empty                                                                                                                                                      |

## Operations – part 1

|               |                                      |
|---------------|--------------------------------------|
| Segment name: | HY72_AG_HD_001_A<br>HY72_AG_HD_001_D |
|---------------|--------------------------------------|

The operation segment includes the operation-based specifications for production and relevant data.

| Field  | V  | T    | L  | D | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | From <sup>7</sup> | To <sup>8</sup> |
|--------|----|------|----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------|
| ANR    | S  | CHAR | 40 |   | Combined order/OP number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1                 | 40              |
| AUART  | K  | CHAR | 5  |   | Order type according to the configuration in HYDRA.<br>The order type must be the same as in the order header and be configured in HYDRA.<br>The attempt of transferring an order type that deviates from the order header is ignored.<br>If this field remains empty, the order type specified in the header is used .                                                                                                                                                                                                                                                                                                               | 41                | 45              |
| AGBEZ  | M  | CHAR | 40 |   | Operation designation/name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 46                | 85              |
| ATK    | K  | CHAR | 40 |   | Article/item (number); HYDRA only displays up to 25 digits/characters.<br>Alphabetic characters in CAPITAL LETTERS<br>If no item/article number is transferred when creating an operation (blank), the article number of the order header will be used for the operation.<br>If no item/article number is transferred when an operation is changed, the article number of the order header will be used.                                                                                                                                                                                                                              | 86                | 125             |
| ATKBEZ | K  | CHAR | 40 |   | Item/article designation/name; HYDRA only displays up to 25 digits.<br>If no item/article designation/name is transferred when an operation is created (blank), the article designation/name of the order header will be used for the operation (redundant). It is recommended to leave this field blank.<br>If no item/article designation/name is transferred when changing an operation, the item/article designation/name of the order header is used.<br>Please note: the item/article designation/name of the order header can only be edited in the MOC; it is automatically used for all operations of the order when saving. | 126               | 165             |
| MATTYP | MM | CHAR | 10 |   | Material type of the item/article;<br>when HYDRA MPL is used, according to the material type configuration<br>If no material type is transferred when creating an operation, the material type of the order header is used.<br>If no material type is transferred when changing an operation, the material type of the order header is used.                                                                                                                                                                                                                                                                                          | 166               | 175             |
| FILLER | K  | CHAR | 10 |   | Reserved; must be blank.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 176               | 185             |

<sup>7</sup> The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA application developers. It does not include the number of characters of the file itself but only of the field SDATA.

<sup>8</sup> See footnote of column "From"

| Field     | V                | T                         | L  | D | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | From                                                                                                                                                                                                                       | To <sup>8</sup> |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
|-----------|------------------|---------------------------|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------|-----|----------|--|-------|------------------|------------------------|-----------|-------|--------------------------|-----------|-----|---------------------------|-----------|-----|---------------------------|-----|-----|
| EXTPRIO   | M                | CHAR                      | 1  |   | Priority (0 - 9; 9 = priority high)<br>If the priority check is enabled for the order type, the priority is always transferred from the order header to the operations. Any deviating priorities of the operation are ignored.                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 186                                                                                                                                                                                                                        | 186             |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| MNR       | M                | CHAR                      | 8  |   | Planned workplace<br>At least one of the fields MNR and/or MGRP must be transferred.<br>When transferring the workplace, HYDRA identifies the workplace's group according to the configuration to avoid inconsistent data. The transferred group is then ignored.                                                                                                                                                                                                                                                                                                                                                                                                                 | 187                                                                                                                                                                                                                        | 194             |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| MGRP      | M                | CHAR                      | 8  |   | Planned group and/or group of the workplace<br>At least one of the fields MNR and/or MGRP must be transferred.<br>If no machine group is transferred, the system automatically identifies the machine group using the transferred workplace.                                                                                                                                                                                                                                                                                                                                                                                                                                      | 195                                                                                                                                                                                                                        | 202             |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| OPT:PLAN  | M                | CHAR                      | 1  |   | Planned<br>M - operation is planned (in detail) for the workplace<br><br>G - operation is in the pool of (groups) (MNR empty)<br>Depending on the workplace (MNR field), the indicator is set internally according to the following logic:<br><table><tr><th>Workplace</th><th>Planned</th><th>Result</th></tr><tr><th>MNR</th><th>OPT:PLAN</th><th></th></tr><tr><td>blank</td><td>blank/ "M"/"G" →</td><td>OPT:PLAN is set to "G"</td></tr><tr><td>not blank</td><td>blank</td><td>→ OPT:PLAN is set to "M"</td></tr><tr><td>not blank</td><td>"M"</td><td>→ OPT:PLAN remains at "M"</td></tr><tr><td>not blank</td><td>"G"</td><td>→ OPT:PLAN remains at "G"</td></tr></table> | Workplace                                                                                                                                                                                                                  | Planned         | Result | MNR | OPT:PLAN |  | blank | blank/ "M"/"G" → | OPT:PLAN is set to "G" | not blank | blank | → OPT:PLAN is set to "M" | not blank | "M" | → OPT:PLAN remains at "M" | not blank | "G" | → OPT:PLAN remains at "G" | 203 | 203 |
| Workplace | Planned          | Result                    |    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                            |                 |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| MNR       | OPT:PLAN         |                           |    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                            |                 |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| blank     | blank/ "M"/"G" → | OPT:PLAN is set to "G"    |    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                            |                 |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| not blank | blank            | → OPT:PLAN is set to "M"  |    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                            |                 |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| not blank | "M"              | → OPT:PLAN remains at "M" |    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                            |                 |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| not blank | "G"              | → OPT:PLAN remains at "G" |    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                            |                 |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| RES:WNR   | K                | CHAR                      | 40 |   | (Main) tool; see notes (see below)!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 204                                                                                                                                                                                                                        | 243             |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| RES:DNC   | K                | CHAR                      | 40 |   | NC program; see notes (see below)!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 244                                                                                                                                                                                                                        | 283             |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| RES:EMAT  | K                | CHAR                      | 40 |   | (Main) input material; see notes (see below)!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 284                                                                                                                                                                                                                        | 323             |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| COLOR     | K                | CHAR                      | 20 |   | Color of the material                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 324                                                                                                                                                                                                                        | 343             |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| KST       | K                | CHAR                      | 8  |   | Cost center                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 344                                                                                                                                                                                                                        | 351             |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| KART      | K                | CHAR                      | 10 |   | Cost type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 352                                                                                                                                                                                                                        | 361             |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| ASTUFE    | K                | CHAR                      | 1  |   | Authorization level to log in and off the OP (lowest authorization = 1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 362                                                                                                                                                                                                                        | 362             |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| RMNR      | K                | CHAR                      | 10 |   | Confirmation/upload number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 363                                                                                                                                                                                                                        | 372             |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| DATTERMB  | K                | DATE                      | 10 |   | Scheduled start (date)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | If scheduling is run outside HYDRA, the scheduled dates of the operations must be transferred from the ERP system.                                                                                                         | 373             | 382    |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| ZEITERMB  | K                | TIME                      | 5  |   | Scheduled start (time)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                            | 383             | 387    |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| DATTERME  | K                | DATE                      | 10 |   | Scheduled end (date)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                            | 388             | 397    |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
| ZEITERME  | K                | TIME                      | 5  |   | Scheduled end (time)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | If scheduling is performed in HYDRA, these fields are overwritten.<br>In HYDRA Shop Floor Scheduling (HLS), the operations are displayed in the pool of groups of the graphic planning board for the scheduled start date. | 398             | 402    |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |
|           |                  |                           |    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                            |                 |        |     |          |  |       |                  |                        |           |       |                          |           |     |                           |           |     |                           |     |     |

| Field           | V | T    | L  | D | Description                                                                                                                                                                                                                                                                | From | To <sup>8</sup> |
|-----------------|---|------|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------|
| DATFB           | K | DATE | 10 |   | Earliest start (date)                                                                                                                                                                                                                                                      | 403  | 412             |
| ZEIFB           | K | TIME | 5  |   | Earliest start (time)                                                                                                                                                                                                                                                      | 413  | 417             |
| DATSB           | K | DATE | 10 |   | Latest start (date)                                                                                                                                                                                                                                                        | 418  | 427             |
| ZEISB           | K | TIME | 5  |   | Latest start (time)                                                                                                                                                                                                                                                        | 428  | 432             |
| DATFE           | K | DATE | 10 |   | Earliest end (date)                                                                                                                                                                                                                                                        | 433  | 442             |
| ZEIFE           | K | TIME | 5  |   | Earliest end (time)                                                                                                                                                                                                                                                        | 443  | 447             |
| DATSE           | K | DATE | 10 |   | Latest end (date)                                                                                                                                                                                                                                                          | 448  | 457             |
| ZEISE           | K | TIME | 5  |   | Latest end (time)                                                                                                                                                                                                                                                          | 458  | 462             |
| DATB            | K | DATE | 10 |   | Planned start (date)                                                                                                                                                                                                                                                       | 463  | 472             |
| ZEIB            | K | TIME | 5  |   | Planned start (time)                                                                                                                                                                                                                                                       | 473  | 477             |
| DATE            | K | DATE | 10 |   | Planned end (date)                                                                                                                                                                                                                                                         | 478  | 487             |
| ZEIE            | K | TIME | 5  |   | Planned end (time)                                                                                                                                                                                                                                                         | 488  | 492             |
| SGR:GUTB        | K | DEC  | 13 | 3 | Target quantity (base quantity unit)                                                                                                                                                                                                                                       | 493  | 507             |
| SGR:GUTP        | M | DEC  | 13 | 3 | Target quantity (primary quantity unit)                                                                                                                                                                                                                                    | 508  | 522             |
| SGR:GUTS        | K | DEC  | 13 | 3 | Target quantity (secondary quantity unit)                                                                                                                                                                                                                                  | 523  | 537             |
| SGR:GUTT        | K | DEC  | 13 | 3 | Target quantity (tertiary quantity unit)                                                                                                                                                                                                                                   | 538  | 552             |
| SGR:AUSB        | K | DEC  | 13 | 3 | Target scrap quantity (base quantity unit)                                                                                                                                                                                                                                 | 553  | 567             |
| SGR:AUSP        | K | DEC  | 13 | 3 | Target scrap quantity (primary quantity unit)                                                                                                                                                                                                                              | 568  | 582             |
| SGR:AUSS        | K | DEC  | 13 | 3 | Target scrap quantity (secondary quantity unit)                                                                                                                                                                                                                            | 583  | 597             |
| SGR:AUST        | K | DEC  | 13 | 3 | Target scrap quantity (tertiary quantity unit)                                                                                                                                                                                                                             | 598  | 612             |
| SGE:B           | K | CHAR | 3  |   | Base quantity unit<br>If no base quantity unit is transferred when creating an operation, the unit of the order header is used, if applicable.<br>If no base quantity unit is transferred when changing an operation, the unit of the order header is used, if applicable. | 613  | 615             |
| SGE:P           | M | CHAR | 3  |   | Primary input quantity unit                                                                                                                                                                                                                                                | 616  | 618             |
| SGE:S           | K | CHAR | 3  |   | Secondary input quantity unit                                                                                                                                                                                                                                              | 619  | 621             |
| SGE:T           | K | CHAR | 3  |   | Tertiary input quantity unit                                                                                                                                                                                                                                               | 622  | 624             |
| WEIGMENGE       | K | DEC  | 13 | 3 | Minimum send-ahead quantity (primary quantity unit)                                                                                                                                                                                                                        | 625  | 639             |
| MENGEPROZ:UNTLL | K | DEC  | 13 | 3 | Underdelivery in percent<br>The value entered is an absolute percentage value of the target quantity (primary quantity unit).<br>Example:<br>- Target quantity of the operation: 120 pieces<br>- Underdelivery: 84%<br>The actual quantity must not fall below 101 items.  | 640  | 654             |



| Field           | V | T    | L  | D | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | From | To <sup>8</sup> |
|-----------------|---|------|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------|
| OPT:UNTLI       | K | CHAR | 1  |   | Reaction to underdelivery<br>Possible values:<br>" " no reaction<br>"W" Warning; entry of a deviation reason required<br>"X" Error; underdelivery not allowed.<br>Note: You can only enter deviation reasons for overdeliveries/ underdeliveries via the CTWIN software. If you use DOS terminals, the reaction "W" is interpreted as an error ("X").                                                                                                                                                                                                                      | 655  | 655             |
| MENGEPROZ:UEBLI | K | DEC  | 13 | 3 | Overdelivery in percent<br>The value entered is an absolute percentage value of the target quantity (primary quantity unit).<br>Example:<br>- Target quantity of the operation: 120 pieces<br>- Overdelivery: 168%<br>The actual quantity must not exceed 201 items.                                                                                                                                                                                                                                                                                                       | 656  | 670             |
| OPT:UEBLI       | K | CHAR | 1  |   | Reaction to overdelivery<br>Possible values:<br>" " no reaction<br>"W" Warning; entry of a deviation reason required<br>"X" Error; overdelivery not allowed.<br>Note: You can only enter deviation reasons for overdeliveries/ underdeliveries via the CTWIN software. If you use DOS terminals, the reaction "W" is interpreted as an error ("X").                                                                                                                                                                                                                        | 671  | 671             |
| UMRFAKTP:Z      | K | NUM  | 8  |   | Numerator for the conversion of primary quantity to basic quantity<br><br>Example:<br>- Base quantity unit: square meter M2<br>- Primary quantity unit: piece ST<br>- 1 piece = 2 square meters.<br>In this case to be stored as<br>- numerator (= UMRFAKTP:Z ) 2 and<br>- denominator (= UMRFAKTP:N) 1.<br><br>MPL for coil-based manufacturing: the conversion is based on specific conversion factors (see the RF structure below); therefore 0 must be transferred for operations that are produced at machines for coil-based production or at coil cutting machines. | 672  | 679             |
| UMRFAKTP:N      | K | NUM  | 8  |   | Denominator for the conversion of primary quantity to basic quantity<br>MPL for coil-based manufacturing: see the note in the UMRFAKTP:Z field                                                                                                                                                                                                                                                                                                                                                                                                                             | 680  | 687             |
| UMRFAKTS:Z      | K | NUM  | 8  |   | Numerator for the conversion of secondary quantity to basic quantity<br>MPL for coil-based manufacturing: see the note in the UMRFAKTP:Z field                                                                                                                                                                                                                                                                                                                                                                                                                             | 688  | 695             |
| UMRFAKTS:N      | K | NUM  | 8  |   | Denominator for the conversion of secondary quantity to basic quantity<br>MPL for coil-based manufacturing: see the note in the UMRFAKTP:Z field                                                                                                                                                                                                                                                                                                                                                                                                                           | 696  | 703             |
| UMRFAKTT:Z      | K | NUM  | 8  |   | Numerator for the conversion of tertiary quantity to basic quantity<br>MPL for coil-based manufacturing: see the note in the UMRFAKTP:Z field                                                                                                                                                                                                                                                                                                                                                                                                                              | 704  | 711             |
| UMRFAKTT:N      | K | NUM  | 8  |   | Denominator for the conversion of tertiary quantity to basic quantity<br>MPL for coil-based manufacturing: see the note in the UMRFAKTP:Z field                                                                                                                                                                                                                                                                                                                                                                                                                            | 712  | 719             |

| Field        | V  | T    | L  | D | Description                                                                                                                                                                                                                                                                         | From | To <sup>8</sup> |
|--------------|----|------|----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------|
| RUEZ         | K  | NUM  | 8  |   | Setup time in seconds. If no setup time exists the value is explicitly to be set to 0.                                                                                                                                                                                              | 720  | 727             |
| RUEZ:ZUSCHL  | K  | NUM  | 8  |   | Additional setup time in seconds. Should be set to 0 if not available.                                                                                                                                                                                                              | 728  | 735             |
| BEARBZEI     | K  | NUM  | 8  |   | Processing time in seconds. Should be set to 0 if not available.                                                                                                                                                                                                                    | 736  | 743             |
| PZ           | K  | NUM  | 8  |   | Inspection time in seconds. Should be set to 0 if not available.                                                                                                                                                                                                                    | 744  | 751             |
| ABRZ         | K  | NUM  | 8  |   | Teardown/retooling time in seconds. Should be set to 0 if not available.                                                                                                                                                                                                            | 752  | 759             |
| LIZ          | K  | NUM  | 8  |   | Delivery time in seconds<br>At present only relevant in connection with the HYDRA Shop Floor Scheduling module (HLS). Should be set to 0 if it is not an external processing OP.                                                                                                    | 760  | 767             |
| FREMDFERT    | K  | CHAR | 1  |   | External processing OP Y/N<br>At present only relevant in connection with the HYDRA Shop Floor Scheduling module (HLS). Should in general be set to "N".                                                                                                                            | 768  | 768             |
| RLZ:EXPR     | K  | CHAR | 6  |   | Remaining run time (formula 1);<br>Relevant if the HYDRA Shop Floor Scheduling module (HLS) and specific BDE applications are used. Generally, transfer "RLFZ" by default. Configure any deviating remaining run time formulas in the system's <a href="#">Formula management</a> . | 769  | 774             |
| RLZ2:EXPR    | K  | CHAR | 6  |   | Remaining run time (formula 2); option (leave empty)                                                                                                                                                                                                                                | 775  | 780             |
| VLZ          | K  | NUM  | 8  |   | Lead time in seconds. Should be set to 0 if not available.                                                                                                                                                                                                                          | 781  | 788             |
| LIEZ:MAX     | K  | NUM  | 8  |   | Max. synchronization time in seconds. Should be set to 0 if not available.                                                                                                                                                                                                          | 789  | 796             |
| WARTZ        | K  | NUM  | 8  |   | Waiting time in seconds. Should be set to 0 if not available.                                                                                                                                                                                                                       | 797  | 804             |
| WARTZ:MIN    | K  | NUM  | 8  |   | Minimum waiting time in seconds. Should be set to 0 if not available.                                                                                                                                                                                                               | 805  | 812             |
| LIEZ         | K  | NUM  | 8  |   | Idle period in seconds. Should be set to 0 if not available.                                                                                                                                                                                                                        | 813  | 820             |
| LART         | K  | CHAR | 4  |   | Wage type                                                                                                                                                                                                                                                                           | 821  | 824             |
| AKKORD       | K  | CHAR | 1  |   | Piecework indicator/ premium                                                                                                                                                                                                                                                        | 825  | 825             |
| TE           | K  | DEC  | 13 | 3 | Premium specification te in seconds per 1000 pieces. Should be set to 0 if not available.                                                                                                                                                                                           | 826  | 840             |
| TR           | K  | DEC  | 13 | 3 | Premium specification tr in seconds. Should be set to 0 if not available.                                                                                                                                                                                                           | 841  | 855             |
| TEB          | K  | DEC  | 13 | 3 | Premium specification teb in seconds per 1000 pieces. Should be set to 0 if not available.                                                                                                                                                                                          | 856  | 870             |
| TRB          | K  | DEC  | 13 | 3 | Premium specification trb in seconds. Should be set to 0 if not available.                                                                                                                                                                                                          | 871  | 885             |
| VERARBCODE   | M  | CHAR | 6  |   | Processing code; "SYSTEM" by default. You can configure additional processing codes in the system.                                                                                                                                                                                  | 886  | 891             |
| OPT:ERF      | M  | CHAR | 1  |   | Recordable true/false                                                                                                                                                                                                                                                               | 892  | 892             |
| OPT:MULTIMNR | M  | CHAR | 1  |   | Parallel production true/false                                                                                                                                                                                                                                                      | 893  | 893             |
| OPT:CNR      | MM | CHAR | 1  |   | Batch management requirement true/false ("true" only relevant for MPL / TRT and/or AIP-MTR)                                                                                                                                                                                         | 894  | 894             |

| Field       | V  | T    | L  | D | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | From | To <sup>8</sup> |
|-------------|----|------|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------|
| OPT:SNR     | MM | CHAR | 1  |   | Serial numbers required (only relevant if serial numbers are used)<br>"G" = Automatic generation of the serial number (only in combination with MPL as batch number)<br>"S" = Automatic generation of the serial number using number range (only in combination with MPL as batch number)<br>"E" = Manual entry of the serial number<br>" " or "N" = serial numbers are not used<br>For details on the implementation/configuration of serial numbers, see <a href="#">here</a> . | 895  | 895             |
| SZY         | K  | NUM  | 8  |   | Target cycle in seconds/ 1000; should be set; mandatory for MDE cycle monitoring                                                                                                                                                                                                                                                                                                                                                                                                  | 896  | 903             |
| TLG         | K  | NUM  | 8  |   | Partitioning; should be pre-populated with 1                                                                                                                                                                                                                                                                                                                                                                                                                                      | 904  | 911             |
| IMPFKT      | K  | DEC  | 13 | 3 | Pulse factor; only integer values allowed! Should be pre-populated with 1 by default.                                                                                                                                                                                                                                                                                                                                                                                             | 912  | 926             |
| OPT:SPLIT   | K  | CHAR | 1  |   | May be split V/N<br>V = Yes, operation may be split<br>N = No, operation must not be split<br>Please note:<br>V only relevant for BDE-SSG, ADE-SPL, HLS-AGS, N must be transferred in the other cases.                                                                                                                                                                                                                                                                            | 927  | 927             |
| MAXANZSPLIT | K  | NUM  | 8  |   | Max. no. of splits. (only relevant if OPT:SPLIT = V)                                                                                                                                                                                                                                                                                                                                                                                                                              | 928  | 935             |
| MBVERH:RUE  | K  | DEC  | 5  | 2 | Machine-operator relation: setup/ PEP workforce requirements: setup                                                                                                                                                                                                                                                                                                                                                                                                               | 936  | 942             |
| MBVERH:NORM | K  | DEC  | 5  | 2 | Machine-operator relation manufacturing/ PEP workforce requirements: manufacturing                                                                                                                                                                                                                                                                                                                                                                                                | 943  | 949             |
| QUAL:NORM   | K  | NUM  | 8  |   | PEP: Qualification: manufacturing                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 950  | 957             |
| QUAL:RUE    | K  | NUM  | 8  |   | PEP: Qualification: setup                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 958  | 965             |



### Fields RES:WNR, RES:DNC, RES:EMAT

The fields

- tool (RES:WNR)
- NC program (RES:DNC)
- input material (RES:EMAT)

are displayed in different evaluation dialogs. Note the following if you populate these fields via the ERP interface:

Only set the operation fields if NO material components are transferred using the structure HY72\_AG\_KOMPL\_001 and NO production resources and tools are transferred using the structure HY72\_AG\_FHM\_001.

If an existing operation is changed, all entries of the component list and of the production resources and tools will automatically be deleted and must then explicitly be transferred.

If a material component changes (change of a material component) or if a production resource and tool is changed (change of a tool), the ERP system must transfer the changed operation as well as the complete material list and the complete list of the

production resources and tools to HYDRA (no "delta download").



### Use of the HYDRA shop floor scheduling system (HLS) or of the graphic order sequencing (BDE-GAV)

If the HYDRA shop floor scheduling (HLS) or the graphic order sequencing (BDE-GAV) are used, you should consider the following issues for operations that are planned with these applications:

You may only transfer operations planned for groups (OPT:PLAN= G). In this case, the planner explicitly carries out the detailed planning (scheduling for the workplace).

If you want to transfer and display operations as planned for a workplace, it is mandatory to populate the following fields:

MNR=<Workplace where the operation must be planned>

PLAN:OPT=M

DATB=<Planned start, date>

ZEIB=<Planned start, time>

DATE=<Planned end, date>

ZEIE=<Planned end, time>

## Operations – part 2

|               |                  |
|---------------|------------------|
| Segment name: | HY72_AG_HD_002_A |
|---------------|------------------|

The segment for the operation includes further operation-based specifications for production and relevant data.

| Field  | V | T    | L  | D | Description                                                                                                       | From <sup>9</sup> | To <sup>10</sup> |
|--------|---|------|----|---|-------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| ANR    | S | CHAR | 40 |   | Combined order/OP number                                                                                          | 1                 | 40               |
| ATKIDX | K | CHAR | 50 |   | Order information → OP → Drawing issue number<br>This field is available as of BDE82 and MLE version HY72PPS_023. | 41                | 90               |

<sup>9</sup> The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA application developers. It does not include the number of characters of the file itself but only of the field SDATA.

<sup>10</sup> See footnote of column "From"

## Long texts of the operation

|               |                                                |
|---------------|------------------------------------------------|
| Segment name: | HY72_AG_INFO_AI_001_A<br>HY72_AG_INFO_AI_001_D |
|---------------|------------------------------------------------|

Use the following structure to transfer text fields as additional information on the operation to HYDRA. The texts are then displayed in the MOC. Each data record contains one page of a maximum of 10 lines and 80 characters of text information each.

The long text of an operation is shown, e.g. in:

- the MOC application "edit operation"
- the MOC application "order information"
- on the AIP shop floor terminal

| Field    | V | T    | L  | D | Description                                                                                                         |
|----------|---|------|----|---|---------------------------------------------------------------------------------------------------------------------|
| KEY      | S | CHAR | 40 |   | Combined order/OP number                                                                                            |
| TYP      | S | CHAR | 2  |   | Record type; by default: "AI"                                                                                       |
| SUBKEY:1 | S | NUM  | 8  |   | Reserved; "00000000" by default.                                                                                    |
| SUBKEY:2 | M | NUM  | 8  |   | Consecutive numbering within the key starting at "00000001".                                                        |
| INFO:BEZ | K | CHAR | 20 |   | Short text; only relevant for SUBKEY:2 = "00000001".<br>If empty, the first 20 digits from Info text 1 will be used |
| INFO:1   | K | CHAR | 80 |   | Info text 1                                                                                                         |
| INFO:2   | K | CHAR | 80 |   | Info text 2                                                                                                         |
| INFO:3   | K | CHAR | 80 |   | Info text 3                                                                                                         |
| INFO:4   | K | CHAR | 80 |   | Info text 4                                                                                                         |
| INFO:5   | K | CHAR | 80 |   | Info text 5                                                                                                         |
| INFO:6   | K | CHAR | 80 |   | Info text 6                                                                                                         |
| INFO:7   | K | CHAR | 80 |   | Info text 7                                                                                                         |
| INFO:8   | K | CHAR | 80 |   | Info text 8                                                                                                         |
| INFO:9   | K | CHAR | 80 |   | Info text 9                                                                                                         |
| INFO:10  | K | CHAR | 80 |   | Info text 10                                                                                                        |

## Components – Component data

|                          |                                            |
|--------------------------|--------------------------------------------|
| Segment name (OBSOLETE): | HY72_AG_KOMPL_001_A<br>HY72_AG_KOMPL_001_D |
|--------------------------|--------------------------------------------|

|                                     |                                            |
|-------------------------------------|--------------------------------------------|
| Segment name:                       | HY72_AG_KOMPL_002_A<br>HY72_AG_KOMPL_002_D |
| Segment name in compatibility mode: | HY72_AG_KOMPL_002                          |

| Field    | V  | T    | L  | D | Description                                                                                                                                                                                                                                                                                                                                       | From <sup>11</sup> | To <sup>12</sup> |
|----------|----|------|----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------|
| ANR      | S  | CHAR | 40 |   | Combined order/OP number                                                                                                                                                                                                                                                                                                                          | 1                  | 40               |
| ATK      | S  | CHAR | 40 |   | Material number;<br>alphabetic characters in CAPITAL LETTERS                                                                                                                                                                                                                                                                                      | 41                 | 80               |
| ATKBEZ   | M  | CHAR | 40 |   | Material designation/name                                                                                                                                                                                                                                                                                                                         | 81                 | 120              |
| BEZ      | K  | CHAR | 30 |   | Comment 1                                                                                                                                                                                                                                                                                                                                         | 121                | 150              |
| BEZ:2    | K  | CHAR | 30 |   | Comment 2                                                                                                                                                                                                                                                                                                                                         | 151                | 180              |
| SLP      | S  | CHAR | 10 |   | BOM item<br>MPL for coil-based manufacturing: item of the component in the layer structure.<br>Each component must have a unique BOM item if several components are used in one operation. Two components must not have the same BOM item.                                                                                                        | 181                | 190              |
| SLS      | M  | NUM  | 8  |   | BOM level<br>Material components with the BOM level > 1 will always be saved under the material type "I" = info component.<br>If you log on input batches via material management (MPL/TRT), you can only log on components of BOM level 0.                                                                                                       | 191                | 198              |
| ART      | MM | CHAR | 2  |   | Material type:<br>"M" (Consumption) material<br>"T" Carrier material (only MPL-RF)<br>"A" Waste component (only MPL-RF)<br>"Z" Additional material (only MPL-RF)<br>"I" Info component                                                                                                                                                            | 199                | 200              |
| MATTYP   | MM | CHAR | 10 |   | MPL: Material type<br>If HYDRA MPL is used, enter a valid material type configured in HYDRA.<br>otherwise: "SYSTEM"                                                                                                                                                                                                                               | 201                | 210              |
| VERBR    | MM | CHAR | 1  |   | MPL: Consumption type<br>If not specified otherwise, assign "L" to this field.<br>For components of the material type "I" assign "N" to the field.<br>For discrete consumption, assign "D" to the field.                                                                                                                                          | 211                | 211              |
| OPT:ERSB | MM | CHAR | 1  |   | MPL-RF: Replaceable – is it possible to use another material than the planned one for such a component? You may only use the same material type that matches the type of the material to be produced.<br>You may only log unplanned material on to the machine, if the respective status is used.<br>MPL: true/false<br>otherwise: "N" by default | 212                | 212              |

<sup>11</sup> The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA application developers. It does not include the number of characters of the file itself but only of the field SDATA.

<sup>12</sup> See footnote of column "From"

| Field        | V  | T    | L  | D | Description                                                                                                                                                                                                                                                                                                                                                         | From <sup>11</sup> | To <sup>12</sup> |
|--------------|----|------|----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------|
| OPT:WZW      | MM | CHAR | 1  |   | MPL: Change necessary; an input batch change for a batch of this material requires an output batch change:<br>if ART = "T" or "Z" -> OPT:WZW must be "J"<br>if ART = "I" or "A" -> OPT:WZW must be "N"<br>if ART = "M" -> OPT:WZW: "J" or "N"                                                                                                                       | 213                | 213              |
| SGR:GUT      | MM | DEC  | 13 | 3 | MPL: Input quantity to produce 1 article in primary quantity unit of the operation.                                                                                                                                                                                                                                                                                 | 214                | 228              |
| SGE:GUT      | MM | CHAR | 3  |   | MPL: Quantity unit of the input quantity                                                                                                                                                                                                                                                                                                                            | 229                | 231              |
| MENGEPROZ    | K  | DEC  | 13 | 3 | Input quantity in percent<br>Reserved; currently not used.                                                                                                                                                                                                                                                                                                          | 232                | 246              |
| UTL          | K  | DEC  | 13 | 3 | Upper tolerance limit in percent; 3 decimal places<br>Reserved; currently not used.                                                                                                                                                                                                                                                                                 | 247                | 261              |
| LTL          | K  | DEC  | 13 | 3 | Lower tolerance limit in percent; 3 decimal places<br>Reserved; currently not used.                                                                                                                                                                                                                                                                                 | 262                | 276              |
| EGR:GUT      | K  | DEC  | 13 | 3 | Total quantity/required quantity: total quantity required for the OP, i.e. for the quantity to be produced (output quantity)                                                                                                                                                                                                                                        | 277                | 291              |
| EGE:GUT      | K  | CHAR | 3  |   | Unit of the required quantity                                                                                                                                                                                                                                                                                                                                       | 292                | 294              |
| SLS:M        | K  | NUMC | 8  |   | BOM level of the parent material<br>Reserved; currently not used.                                                                                                                                                                                                                                                                                                   | 295                | 302              |
| SLP:M        | K  | CHAR | 10 |   | BOM item of the parent material<br>Reserved; currently not used.                                                                                                                                                                                                                                                                                                    | 303                | 312              |
| MENGE:FIX    | K  | CHAR | 1  |   | Indicator: Fixed quantity<br>Reserved; currently not used.                                                                                                                                                                                                                                                                                                          | 313                | 313              |
| PPS:RETRO    | K  | CHAR | 1  |   | Indicator: Backflush (retrograde withdrawal) (in ERP)<br>Reserved; currently not used.                                                                                                                                                                                                                                                                              | 314                | 314              |
| MENGETOL     | K  | DEC  | 13 | 3 | Tolerance in percent<br>Please note: only in segment HY72_AG_KOMPL_002_A                                                                                                                                                                                                                                                                                            | 315                | 329              |
| MENGEABW     | K  | DEC  | 13 | 3 | Deviation in percent<br>Please note: only in segment HY72_AG_KOMPL_002_A                                                                                                                                                                                                                                                                                            | 330                | 344              |
| OPT:SLOS     | K  | CHAR | 1  |   | Flag for merged batch<br>As of MPL82 / MLE version HY72PPS_019<br>Identification of the component as part of the collection of serial numbers for merging<br>" " If no serial numbers are collected<br>and<br>if serial numbers are collected and merged:<br>subordinated serial number<br>"F" if serial numbers are collected and merged:<br>leading serial number | 345                | 345              |
| VERB:ZAEHLER | K  | CHAR | 1  |   | Indicator for automatic counter consumption (as of MPL82 / MLE version HY72PPS_019):<br>"J" Automatic counter consumption enabled<br>"N" Automatic counter consumption <i>disabled</i>                                                                                                                                                                              | 346                | 346              |

If a material component changes (change of a material component) or if a production resource and tool is changed (change of a tool), the ERP system must transfer the changed operation as well as the complete material list and the complete list of the production resources and tools to HYDRA (no "delta download").



Also see the notes in the "Operations" chapter.

During the data transfer to HYDRA, the first material transferred to HYDRA will be displayed in the operation.

## Component - user fields

|               |                            |
|---------------|----------------------------|
| Segment name: | HY72_AG_KOMPL_USRFLD_001_A |
|---------------|----------------------------|

Use user fields to store further customer-specific information to HYDRA in addition to the fields that are available by default. Use this segment to transfer this data from the ERP system to HYDRA and to store this data in the component.

The so-called user field key specifies the available user fields and their meaning. Each user field key describes a combination of user fields. The document [Configuration Userfields.pdf](#) describes how to configure the user field key.



To allow for a consistent data exchange between the ERP system and HYDRA, the customer must ensure that the user field keys are identical in both systems.

| Field  | V | T    | L  | D | Description                                                                                    | From <sup>13</sup> | To <sup>14</sup> |
|--------|---|------|----|---|------------------------------------------------------------------------------------------------|--------------------|------------------|
| ANR    | S | CHAR | 40 |   | Combined order/OP number                                                                       | 1                  | 40               |
| ATK    | S | CHAR | 40 |   | Material number;<br>alphabetic characters in CAPITAL LETTERS                                   | 41                 | 80               |
| SLP    | S | CHAR | 10 |   | BOM item<br>MPL for coil-based manufacturing: item of the component<br>in the layer structure. | 81                 | 90               |
| USRFLD | S | CHAR | 8  |   | User field key. The user field key must be configured in the<br>system.                        | 91                 | 98               |

<sup>13</sup> The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA application developers. It does not include the number of characters of the file itself but only of the field SDATA.

<sup>14</sup> See footnote of column "From"



| Field | V | T    | L  | D | Description   | From<br>13 | To <sup>14</sup> |
|-------|---|------|----|---|---------------|------------|------------------|
| FU:1  | K | DATE | 10 |   | User field 1  | 99         | 108              |
| FU:2  | K | DATE | 10 |   | User field 2  | 109        | 118              |
| FU:3  | K | DATE | 10 |   | User field 3  | 119        | 128              |
| FU:4  | K | DATE | 10 |   | User field 4  | 129        | 138              |
| FU:5  | K | DATE | 10 |   | User field 5  | 139        | 148              |
| FU:6  | K | DATE | 10 |   | User field 6  | 149        | 158              |
| FU:7  | K | NUM  | 8  |   | User field 7  | 159        | 166              |
| FU:8  | K | NUM  | 8  |   | User field 8  | 167        | 174              |
| FU:9  | K | NUM  | 8  |   | User field 9  | 175        | 182              |
| FU:10 | K | NUM  | 8  |   | User field 10 | 183        | 190              |
| FU:11 | K | NUM  | 8  |   | User field 11 | 191        | 198              |
| FU:12 | K | NUM  | 8  |   | User field 12 | 199        | 206              |
| FU:13 | K | NUM  | 8  |   | User field 13 | 207        | 214              |
| FU:14 | K | NUM  | 8  |   | User field 14 | 215        | 222              |
| FU:15 | K | NUM  | 8  |   | User field 15 | 223        | 230              |
| FU:16 | K | NUM  | 8  |   | User field 16 | 231        | 238              |
| FU:17 | K | NUM  | 8  |   | User field 17 | 239        | 246              |
| FU:18 | K | NUM  | 8  |   | User field 18 | 247        | 254              |
| FU:19 | K | NUM  | 8  |   | User field 19 | 255        | 262              |
| FU:20 | K | NUM  | 8  |   | User field 20 | 263        | 270              |
| FU:21 | K | NUM  | 8  |   | User field 21 | 271        | 278              |
| FU:22 | K | NUM  | 8  |   | User field 22 | 279        | 286              |
| FU:23 | K | DEC  | 13 | 3 | User field 23 | 287        | 301              |
| FU:24 | K | DEC  | 13 | 3 | User field 24 | 302        | 316              |
| FU:25 | K | DEC  | 13 | 3 | User field 25 | 317        | 331              |
| FU:26 | K | DEC  | 13 | 3 | User field 26 | 332        | 346              |
| FU:27 | K | DEC  | 13 | 3 | User field 27 | 347        | 361              |
| FU:28 | K | DEC  | 13 | 3 | User field 28 | 362        | 376              |
| FU:29 | K | CHAR | 1  |   | User field 29 | 377        | 377              |
| FU:30 | K | CHAR | 1  |   | User field 30 | 378        | 378              |
| FU:31 | K | CHAR | 1  |   | User field 31 | 379        | 379              |
| FU:32 | K | CHAR | 1  |   | User field 32 | 380        | 380              |
| FU:33 | K | CHAR | 1  |   | User field 33 | 381        | 381              |
| FU:34 | K | CHAR | 1  |   | User field 34 | 382        | 382              |
| FU:35 | K | CHAR | 1  |   | User field 35 | 383        | 383              |

| Field | V | T    | L  | D | Description   | From<br>13 | To <sup>14</sup> |
|-------|---|------|----|---|---------------|------------|------------------|
| FU:36 | K | CHAR | 1  |   | User field 36 | 384        | 384              |
| FU:37 | K | CHAR | 1  |   | User field 37 | 385        | 385              |
| FU:38 | K | CHAR | 1  |   | User field 38 | 386        | 386              |
| FU:39 | K | CHAR | 1  |   | User field 39 | 387        | 387              |
| FU:40 | K | CHAR | 1  |   | User field 40 | 388        | 388              |
| FU:41 | K | CHAR | 1  |   | User field 41 | 389        | 389              |
| FU:42 | K | CHAR | 1  |   | User field 42 | 390        | 390              |
| FU:43 | K | CHAR | 1  |   | User field 43 | 391        | 391              |
| FU:44 | K | CHAR | 1  |   | User field 44 | 392        | 392              |
| FU:45 | K | CHAR | 10 |   | User field 45 | 393        | 402              |
| FU:46 | K | CHAR | 10 |   | User field 46 | 403        | 412              |
| FU:47 | K | CHAR | 10 |   | User field 47 | 413        | 422              |
| FU:48 | K | CHAR | 10 |   | User field 48 | 423        | 432              |
| FU:49 | K | CHAR | 10 |   | User field 49 | 433        | 442              |
| FU:50 | K | CHAR | 10 |   | User field 50 | 443        | 452              |
| FU:51 | K | CHAR | 20 |   | User field 51 | 453        | 472              |
| FU:52 | K | CHAR | 20 |   | User field 52 | 473        | 492              |
| FU:53 | K | CHAR | 20 |   | User field 53 | 493        | 512              |
| FU:54 | K | CHAR | 20 |   | User field 54 | 513        | 532              |
| FU:55 | K | CHAR | 20 |   | User field 55 | 533        | 552              |
| FU:56 | K | CHAR | 20 |   | User field 56 | 553        | 572              |
| FU:57 | K | CHAR | 20 |   | User field 57 | 573        | 592              |
| FU:58 | K | CHAR | 20 |   | User field 58 | 593        | 612              |
| FU:59 | K | CHAR | 20 |   | User field 59 | 613        | 632              |
| FU:60 | K | CHAR | 20 |   | User field 60 | 633        | 652              |
| FU:61 | K | CHAR | 20 |   | User field 61 | 653        | 672              |
| FU:62 | K | CHAR | 20 |   | User field 62 | 673        | 692              |
| FU:63 | K | CHAR | 20 |   | User field 63 | 693        | 712              |
| FU:64 | K | CHAR | 20 |   | User field 64 | 713        | 732              |
| FU:65 | K | CHAR | 40 |   | User field 65 | 733        | 772              |
| FU:66 | K | CHAR | 40 |   | User field 66 | 773        | 812              |

## Production resources and tools

|               |                                        |
|---------------|----------------------------------------|
| Segment name: | HY72_AG_FHM_001_A<br>HY72_AG_FHM_001_D |
|---------------|----------------------------------------|

If you want to use production resources and tools as resources in HYDRA, you also have to use the HYDRA Tool and Resource Management (WRM) and/or of HYDRA DNC. If production resources and tools are only displayed in the BDE module, you do not have to use WRM or DNC.

| Field   | K | T    | L  |   | Description                                                                                                                                                                                                                                                                                                                                                |
|---------|---|------|----|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANR     | S | CHAR | 40 |   | Combined order/OP number                                                                                                                                                                                                                                                                                                                                   |
| RESTYP  | S | CHAR | 4  |   | Resource type; possible values:<br>DNC DNC program<br>ENT removal device<br>TEM temperature device<br>VOR device<br>WNR tool<br><br>If you use the HYDRA tools and resources management (WRM), you can define additional resource types (idents) in the application's resource type configuration (Menu <i>WRM: Master data</i> > <i>Resource types</i> ). |
| ATK     | S | CHAR | 40 |   | Resource/ material number;<br>alphabetic characters in CAPITAL LETTERS                                                                                                                                                                                                                                                                                     |
| ATKBEZ  | K | CHAR | 40 |   | Designation                                                                                                                                                                                                                                                                                                                                                |
| BEZ     | K | CHAR | 30 |   | Comment 1                                                                                                                                                                                                                                                                                                                                                  |
| BEZ:2   | K | CHAR | 30 |   | Comment 2                                                                                                                                                                                                                                                                                                                                                  |
| SGR:GUT | M | DEC  | 13 | 3 | Input quantity<br>Please note: The quantity of a production resource and tool that is to be checked in the HYDRA shop floor scheduling module (license: HLS-BSR) is generally assumed to be 1, if 0 has been transferred here.                                                                                                                             |
| SGE:GUT | K | CHAR | 3  |   | Quantity unit                                                                                                                                                                                                                                                                                                                                              |



If a material component changes (change of a material component) or if a production resource and tool is changed (change of a tool), the ERP system must transfer the changed operation as well as the complete material list and the complete list of the production resources and tools to HYDRA (no "delta download").

Also see the notes in the "Operations" chapter.

During the data transfer to HYDRA, the first production resource and tool transferred with RESTYP=DNC to HYDRA will be taken over in the operation and displayed as NC program.

Otherwise, the first production resource and tool transferred with RESTYP<>DOC to HYDRA will be taken over for the operation and displayed as tool. If production resources and tools of different resource types are transferred, the customer must ensure a useful sequence (e.g. first the prod. resources + tools of the resource type WNR and then the production resources and tools of another resource type).

Use the option "automatic creation" in the [Resource types](#) configuration to make sure that production resources and tools of a resource type are created automatically as resource, provided that they do not yet exist in the system (only relevant if the WRM product group (Tool and Resource Management) is in use).

## Documents

|               |                                        |
|---------------|----------------------------------------|
| Segment name: | HY72_AG_DOC_001_A<br>HY72_AG_DOC_001_D |
|---------------|----------------------------------------|

| Field  | V | T    | L   | D | Description                                                          |
|--------|---|------|-----|---|----------------------------------------------------------------------|
| ANR    | S | CHAR | 40  |   | Combined order/OP number                                             |
| ATK    | S | CHAR | 40  |   | Document ID: unique key alphabetic characters in CAPITAL LETTERS     |
| ATKBEZ | M | CHAR | 40  |   | Designation                                                          |
| BEZ    | K | CHAR | 30  |   | Comment 1                                                            |
| BEZ:2  | K | CHAR | 30  |   | Comment 2                                                            |
| PATH   | M | CHAR | 8   |   | Refers to a path defined in the <a href="#">Path configuration</a> . |
| FILE   | M | CHAR | 128 |   | File name incl. file extension                                       |

## Operation user fields

|               |                      |
|---------------|----------------------|
| Segment name: | HY72_AG_USRFLD_001_A |
|---------------|----------------------|

Use user fields to store further customer-specific information to HYDRA in addition to the fields that are available by default. Use this segment to transfer this data from the ERP system to HYDRA and to store this data in the operation.

The so-called user field key specifies the available user fields and their meaning. Each user field key describes a combination of user fields. The document [Configuration Userfields.pdf](#) describes how to configure the user field key.



To allow for a consistent data exchange between the ERP system and HYDRA, the customer must ensure that the user field keys are identical in both systems.

| Field | V | T    | L  | D | Description              | From | To |
|-------|---|------|----|---|--------------------------|------|----|
| ANR   | S | CHAR | 40 |   | Combined order/OP number | 1    | 40 |

|        |   |      |    |   |                                                                      |     |     |
|--------|---|------|----|---|----------------------------------------------------------------------|-----|-----|
| USRFLD | S | CHAR | 8  |   | User field key. The user field key must be configured in the system. | 41  | 48  |
| FU:1   | K | DATE | 10 |   | User field 1                                                         | 49  | 58  |
| FU:2   | K | DATE | 10 |   | User field 2                                                         | 59  | 68  |
| FU:3   | K | DATE | 10 |   | User field 3                                                         | 69  | 78  |
| FU:4   | K | DATE | 10 |   | User field 4                                                         | 79  | 88  |
| FU:5   | K | DATE | 10 |   | User field 5                                                         | 89  | 98  |
| FU:6   | K | DATE | 10 |   | User field 6                                                         | 99  | 108 |
| FU:7   | K | NUM  | 8  |   | User field 7                                                         | 109 | 116 |
| FU:8   | K | NUM  | 8  |   | User field 8                                                         | 117 | 124 |
| FU:9   | K | NUM  | 8  |   | User field 9                                                         | 125 | 132 |
| FU:10  | K | NUM  | 8  |   | User field 10                                                        | 133 | 140 |
| FU:11  | K | NUM  | 8  |   | User field 11                                                        | 141 | 148 |
| FU:12  | K | NUM  | 8  |   | User field 12                                                        | 149 | 156 |
| FU:13  | K | NUM  | 8  |   | User field 13                                                        | 157 | 164 |
| FU:14  | K | NUM  | 8  |   | User field 14                                                        | 165 | 172 |
| FU:15  | K | NUM  | 8  |   | User field 15                                                        | 173 | 180 |
| FU:16  | K | NUM  | 8  |   | User field 16                                                        | 181 | 188 |
| FU:17  | K | NUM  | 8  |   | User field 17                                                        | 189 | 196 |
| FU:18  | K | NUM  | 8  |   | User field 18                                                        | 197 | 204 |
| FU:19  | K | NUM  | 8  |   | User field 19                                                        | 205 | 212 |
| FU:20  | K | NUM  | 8  |   | User field 20                                                        | 213 | 220 |
| FU:21  | K | NUM  | 8  |   | User field 21                                                        | 221 | 228 |
| FU:22  | K | NUM  | 8  |   | User field 22                                                        | 229 | 236 |
| FU:23  | K | DEC  | 13 | 3 | User field 23                                                        | 237 | 251 |
| FU:24  | K | DEC  | 13 | 3 | User field 24                                                        | 252 | 266 |
| FU:25  | K | DEC  | 13 | 3 | User field 25                                                        | 267 | 281 |
| FU:26  | K | DEC  | 13 | 3 | User field 26                                                        | 282 | 296 |
| FU:27  | K | DEC  | 13 | 3 | User field 27                                                        | 297 | 311 |
| FU:28  | K | DEC  | 13 | 3 | User field 28                                                        | 312 | 326 |
| FU:29  | K | CHAR | 1  |   | User field 29                                                        | 327 | 327 |
| FU:30  | K | CHAR | 1  |   | User field 30                                                        | 328 | 328 |
| FU:31  | K | CHAR | 1  |   | User field 31                                                        | 329 | 329 |
| FU:32  | K | CHAR | 1  |   | User field 32                                                        | 330 | 330 |
| FU:33  | K | CHAR | 1  |   | User field 33                                                        | 331 | 331 |
| FU:34  | K | CHAR | 1  |   | User field 34                                                        | 332 | 332 |
| FU:35  | K | CHAR | 1  |   | User field 35                                                        | 333 | 333 |

|       |   |      |    |  |                                                                                   |     |     |
|-------|---|------|----|--|-----------------------------------------------------------------------------------|-----|-----|
| FU:36 | K | CHAR | 1  |  | User field 36                                                                     | 334 | 334 |
| FU:37 | K | CHAR | 1  |  | User field 37                                                                     | 335 | 335 |
| FU:38 | K | CHAR | 1  |  | User field 38                                                                     | 336 | 336 |
| FU:39 | K | CHAR | 1  |  | User field 39                                                                     | 337 | 337 |
| FU:40 | K | CHAR | 1  |  | User field 40                                                                     | 338 | 338 |
| FU:41 | K | CHAR | 1  |  | User field 41                                                                     | 339 | 339 |
| FU:42 | K | CHAR | 1  |  | User field 42                                                                     | 340 | 340 |
| FU:43 | K | CHAR | 1  |  | User field 43                                                                     | 341 | 341 |
| FU:44 | K | CHAR | 1  |  | User field 44                                                                     | 342 | 342 |
| FU:45 | K | CHAR | 10 |  | User field 45                                                                     | 343 | 352 |
| FU:46 | K | CHAR | 10 |  | User field 46                                                                     | 353 | 362 |
| FU:47 | K | CHAR | 10 |  | User field 47                                                                     | 363 | 372 |
| FU:48 | K | CHAR | 10 |  | User field 48                                                                     | 373 | 382 |
| FU:49 | K | CHAR | 10 |  | User field 49                                                                     | 383 | 392 |
| FU:50 | K | CHAR | 10 |  | User field 50                                                                     | 393 | 402 |
| FU:51 | K | CHAR | 20 |  | User field 51                                                                     | 403 | 422 |
| FU:52 | K | CHAR | 20 |  | User field 52                                                                     | 423 | 442 |
| FU:53 | K | CHAR | 20 |  | User field 53<br>The shop floor terminal shows this field in the Comment 1 field  | 443 | 462 |
| FU:54 | K | CHAR | 20 |  | User field 54<br>The shop floor terminal shows this field in the Comment 2 field. | 463 | 482 |
| FU:55 | K | CHAR | 20 |  | User field 55                                                                     | 483 | 502 |
| FU:56 | K | CHAR | 20 |  | User field 56                                                                     | 503 | 522 |
| FU:57 | K | CHAR | 20 |  | User field 57                                                                     | 523 | 542 |
| FU:58 | K | CHAR | 20 |  | User field 58                                                                     | 543 | 562 |
| FU:59 | K | CHAR | 20 |  | User field 59                                                                     | 563 | 582 |
| FU:60 | K | CHAR | 20 |  | User field 60                                                                     | 583 | 602 |
| FU:61 | K | CHAR | 20 |  | User field 61                                                                     | 603 | 622 |
| FU:62 | K | CHAR | 20 |  | User field 62                                                                     | 623 | 642 |
| FU:63 | K | CHAR | 20 |  | User field 63                                                                     | 643 | 662 |
| FU:64 | K | CHAR | 20 |  | User field 64                                                                     | 663 | 682 |
| FU:65 | K | CHAR | 40 |  | User field 65                                                                     | 683 | 722 |
| FU:66 | K | CHAR | 40 |  | User field 66                                                                     | 723 | 762 |

## Specific data for coil-based production

|               |                  |
|---------------|------------------|
| Segment name: | HY72_AG_RF_001_A |
|---------------|------------------|

This segment depends on the operation, i.e. the data transferred here populate operation-specific fields in HYDRA. You require this data, if you use the MPL module for coil-based manufacturing.

| Field name                                                                                     | V  | T    | L  | D | Description                                                                                                                                                                                                                                                                                                                                                        | From | To |
|------------------------------------------------------------------------------------------------|----|------|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|
| ANR                                                                                            | S  | CHAR | 40 |   | Combined order/OP number                                                                                                                                                                                                                                                                                                                                           | 1    | 40 |
| RFAGTYP       | K  | CHAR | 1  |   | Special indicator; type of the operation:<br>" " No special processing<br>"P" Packing operation; required in combination with the central weighing function.                                                                                                                                                                                                       | 41   | 41 |
| RFABZ         | MM | CHAR | 1  |   | Identifies mother and child OPs for a planned branch.<br>"M" Mother operation of a planned branch<br>"K" Child OP of a planned branch, which will be confirmed/uploaded in a special goods movement (531) at the goods issue.<br>The indicator should only then be set to "K" if the planned branches must really be confirmed/uploaded as special goods movement. | 42   | 42 |
| RFOPT:RS    | MM | CHAR | 1  |   | Indicator roll cutting (only relevant if this is a cutting OP)<br>" " No cutting OP<br>"T" Cutting OP with numbering of daughter rolls (batch number has 15 digits: 10 digits+"-"+4-digit index)<br>"M" Cutting OP with numbering of mother rolls; this leads to further mother rolls (batch number has 10 digits).                                                | 43   | 43 |
| RFMANR      | MM | CHAR | 40 |   | In coil cutting, this reference to the mother OP is used to define the cutting plan.<br>An OP that does <i>not</i> reference to a mother OP <b>must</b> generally refer to itself.<br>Please note: the HYDRA order ID must be configured.                                                                                                                          | 44   | 83 |
| RFTRANZ     | MM | NUMC | 5  |   | In case of cutting operations: Number of planned daughter coils per cut. Corresponds to the planned output batches of the operation.<br>If the cutting plan is not defined, 0 is entered here.                                                                                                                                                                     | 84   | 88 |
| RFTRANZSUM  | MM | NUMC | 5  |   | For cutting operations (mother OP): number of planned daughter coils per cut (encompassing all branched off material). No specific processing in HYDRA.<br>If the cutting plan is not defined, 0 is entered here.                                                                                                                                                  | 89   | 93 |
| RFRAZ       | MM | NUMC | 6  |   | Planned total number of coils to be produced (mother and daughter coils); no specific processing in HYDRA.                                                                                                                                                                                                                                                         | 94   | 99 |

| Field name    | V  | T    | L  | D | Description                                                                                                                                                                                                                                                                                                              | From | To  |
|---------------|----|------|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| RFSTKF        | MM | NUMC | 8  |   | Surface of a piece of the article to be produced.<br>Unit: MM2/ PCE (integer)                                                                                                                                                                                                                                            | 100  | 107 |
| RFBSBRS       | MM | DEC  | 10 | 3 | Total seam width in mm<br>If several coils are produced at the same time in one operation, this field includes the sum total of the separate seam widths.<br>For the planned branches the seam width of the single operation (no summaries) is explicitly set for each operation ("mother" and "child" OPs).<br>Unit: MM | 108  | 119 |
| RFBREITEE     | MM | DEC  | 10 | 3 | Input width of operation<br>Unit: MM                                                                                                                                                                                                                                                                                     | 120  | 131 |
| RFBREITEA     | MM | DEC  | 10 | 3 | Output width of operation<br>Width of <i>one</i> coil within an operation.<br>For the planned branches the initial width of the single operation (no summaries) is explicitly set for each operation ("mother" and "child" OPs).<br>Unit: MM                                                                             | 132  | 143 |
| RFAGVFA       | MM | DEC  | 10 | 3 | Mass per unit area<br>Unit: G/M2                                                                                                                                                                                                                                                                                         | 144  | 155 |
| Casing weight | K  | DEC  | 10 | 3 | Specifies the casing weight of the daughter coils for cutting operations.<br>Unit: G                                                                                                                                                                                                                                     | 156  | 167 |

## Order network

|               |                                      |
|---------------|--------------------------------------|
| Segment name: | HY72_ANETZ_001_A<br>HY72_ANETZ_001_D |
|---------------|--------------------------------------|

This segment is a global segment used for several orders. Use this segment to define global relationships for several orders. These relationships are respected during planning in the HYDRA shop floor scheduling (HLS) and/or during processing (e.g. target quantity update, if enabled) when the affected operations are planned and/or entered. Data can only be transferred if the transferred orders and/or their operations already exist in HYDRA.

| Field         | V | T    | L  | D | Description                                     |
|---------------|---|------|----|---|-------------------------------------------------|
| Preceding OP  | S | CHAR | 40 |   | Combined order/ OP number of the predecessor OP |
| Succeeding OP | S | CHAR | 40 |   | Combined order/ OP number of the successor OP   |
| AOB           | S | CHAR | 2  |   | "ES" by default                                 |



Only end/start relations are supported.

You may not transfer relationships between adjacent operations of an order. Relationships are neither supported for split operations and for merged operations and may therefore not be set.





You need the license HLS-KAN if you want to integrate these relationships in the HYDRA Shop Floor Scheduling module (HLS).

## Production variants

|               |                                          |
|---------------|------------------------------------------|
| Segment name: | HY72_FERTVAR_001_A<br>HY72_FERTVAR_001_D |
|---------------|------------------------------------------|

Use this segment to enter / edit production variants in HYDRA. If configured accordingly , these variants are integrated when identifying production variants as part of the order transfer.

| Field                   | V  | T    | L  | D | Description                                                                       |
|-------------------------|----|------|----|---|-----------------------------------------------------------------------------------|
| Version                 | S  | CHAR | 10 |   | Version                                                                           |
| Status                  | M  | CHAR | 1  |   | Status of the production variant<br>F = Released<br>S = Blocked                   |
| Article                 | S  | CHAR | 40 |   | Article                                                                           |
| Material type           | S  | CHAR | 10 |   | Material type; blank by default (currently not processed)                         |
| Machine                 | S  | CHAR | 8  |   | Machine                                                                           |
| Group                   | S  | CHAR | 8  |   | Group                                                                             |
| Number (machine)        | K  | DEC  | 13 | 3 | Number of machines; "1" by default (currently not processed)                      |
| Resource                | M  | CHAR | 40 |   | Resource.<br>Note: The default resource is always assigned the resource type WNR. |
| Resource family         | K  | CHAR | 18 |   | Resource family (currently not processed)                                         |
| Number (resource)       | K  | NUM  | 8  |   | Number of resources                                                               |
| Target cycle            | K  | NUM  | 8  |   | Target cycle in [seconds/ 1000 cycles]                                            |
| Partitioning            | K  | DEC  | 13 | 3 | Partitioning                                                                      |
| Setup time              | K  | NUM  | 8  |   | Setup time (seconds)                                                              |
| Teardown/retooling time | K  | NUM  | 8  |   | Teardown/retooling time (seconds)                                                 |
| Data ID                 | SA | CHAR | 15 |   | Data ID                                                                           |
| Comment                 | K  | CHAR | 50 |   | Comment                                                                           |
| Priority                | M  | NUMC | 1  |   | Priority                                                                          |
| Valid from              | K  | DATE | 10 |   | Valid from (MM/DD/YYYY)                                                           |
| Valid until             | K  | DATE | 10 |   | Valid until (MM/DD/YYYY)                                                          |

## 6 Uploading data types: HYDRA --> ERP

| Type             | Description                                                                                                                                                                                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CHAR x</b>    | Information is left-aligned for the data type CHAR; unnecessary places are filled with blanks.<br><br>Example: "ABCD "                                                                                                                                                                                                                                  |
| <b>NUM x</b>     | Numeric field of the length x without sign. Numbers are right-aligned; unnecessary places are filled with zeros.<br><br>Example: "00000002"                                                                                                                                                                                                             |
| <b>DEC_O x.y</b> | Numeric field of the length x and y decimal places. An algebraic sign is preceding the data field ("+" or "-"). Places that are not required are filled with zeros. There is NO DECIMAL SEPARATOR.<br><br>e.g. DEC_O 13,3: <ul style="list-style-type: none"> <li>• -1234567890,123 → -1234567890123</li> <li>• 234567890,3 → +0234567890300</li> </ul> |
| <b>DATE</b>      | The date is displayed in the YYYYMMDD format.<br><br>The field is filled with blanks (if it is not required).                                                                                                                                                                                                                                           |
| <b>TIME</b>      | The time is transferred in the HHMMSS format.<br><br>The field is populated with "000000".                                                                                                                                                                                                                                                              |

Generally, HYDRA always transfers a contiguous data structure. Data fields that are not used are filled with blanks. The following definitions apply if you use the file port:

Each data record included in the file has to be completed by 'CR' (U+000D) and 'LF' (U+000A) for Windows and 'LF' (U+000A) for Unix.

HYDRA expects the file to be in the UTF-8 format and HYDRA also uses this format for uploads. On request, the file transfer may also be performed in the file format that was used until MW 2.0.

## 7 Operation-related Uploads HYDRA => ERP

### Overview

|  |    |  |
|--|----|--|
|  | →→ |  |
|  | →  |  |

The system uses an individual IDOC with different interface record types (*SATZART* field in the interface record) to transfer the operation-related uploads recorded in HYDRA to the ERP system. There are two different interface record types:

- Uploads based on operations (*SATZART*/record type = "A")

All data records relating to operations are uploaded using the interface record type "A". HYDRA uses a separate entry record type to manage these data records (see e.g. the MOC application [Order-related postings](#)). This indicator allows for the records assigned to the interface record type "A" to be differentiated in more detail. The following entry record types are uploaded:

- Interruption of an operation (postings of the entry record type "U" (*STEUER* field in the interface record); is also performed at the end of the shift if the automatic shift function is active).
- Completion of an operation (postings of entry record type "E" (*STEUER\_KZ* field in the interface record)).

A data record where quantities, processing duration and RPA are not filled out is transferred for each posting relating to orders (interruption, completion) when it comes to operations that are subject to batch management. These data records only transfer the labor utilization of this order. When finishing the order, the data record indicates that the order has actually been finished.

Quantities and times are uploaded in separate data records that are generated in HYDRA while the output batch is changed.

- Personal upload (*SATZART*/record type = "P")

If HYDRA is customized accordingly, you can also upload the "B" records that are managed in HYDRA for the single persons. To do so, use the interface record type "P".

Data is provided at regular intervals (by default approx. every 60 minutes) in the HYDRA subdirectory *./inf\_int/interf* (standard system). This directory is located in the HYDRA directory or in the system directory in case of a multi-system environment.

Data relevant to operations, such as quantities or times, are posted in the ERP system. The ERP system manages and processes wrong postings separately.

The following specifications result for the IDOC:

|                       |                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Message type:</b>  | HY72ADRCK_TT                                                                                                          |
| <b>IDOC type:</b>     | HY72ADRCK_TT                                                                                                          |
| <b>File name:</b>     | HY72ADRCK_TIMETICKET                                                                                                  |
| <b>File extension</b> | According to configuration in MLE communication (Logical systems > Outbound Configuration File Port)<br>Usually: .dat |



It might be the case that negative yield quantities are posted when finishing the operation, provided that part quantities are reported (requires customization services by MPDV) and data is collected at the same time via the total quantity counter on machines working with HYDRA-MDE. Remove the parameter /NEG\_MENGE from the myerprck.scr script, provided that the ERP system does not intend to post negative quantities.



An "S" (cancellation) entered in the ERFART field indicates corrected postings. You can find this value in the *Input type* field of the maintenance of postings. Provided that corrections are uploaded, the system uploads the canceled values as absolute values (without algebraic signs), i.e. with an "S" in the ERFART field.

## Data structure

| Field  | T    | L | D | Description                                                                                                                                                          | From <sup>15</sup> | To <sup>16</sup> |
|--------|------|---|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------|
| SART   | CHAR | 1 |   | Interface record type<br>"A" uploads based on operations<br>"P" uploads based on persons                                                                             | 1                  | 1                |
| ERFART | CHAR | 1 |   | Origin<br>" " Original data record as recorded<br>"E" data record created manually in the <i>maintenance of postings</i> dialog (edited)<br>"S" Cancellation for ERP | 2                  | 2                |

<sup>15</sup> The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA application developers. It does not include the number of characters of the file itself but only of the field SDATA.

<sup>16</sup> See footnote of column "From"

| Field     | T     | L  | D | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | From<br>15 | To <sup>16</sup> |
|-----------|-------|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|
| RMNR      | CHAR  | 40 |   | Upload/confirmation number (if available in HYDRA)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3          | 42               |
| ANR       | CHAR  | 40 |   | Order number<br>The exact length that is uploaded/confirmed depends on how the lengths are configured for the order or operation in the HYDRA basic parameter settings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 43         | 82               |
| AUART     | CHAR  | 5  |   | Order type of the order; according to HYDRA configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 83         | 87               |
| STEUER_KZ | CHAR  | 1  |   | Control indicator of the operation status when being recorded. The indicator depends on the record type of the ADE log record.<br>"L" OP is running (record type "A")<br>"E" OP is finished (record type "E")<br>"U" OP is interrupted (record type "U", "T", "H")<br>Please note: - If you use split OPs, the system uploads/confirm the status of the split master.- The value of this field is not defined for personal uploads.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 88         | 88               |
| AG_STATUS | CHAR  | 5  |   | Operation status when the ERP upload is being performed, according to the HYDRA configuration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 89         | 93               |
| SCHICHTNR | NUM   | 2  |   | Shift according to the shift model assigned to the work center during which the BDE posting was made. The shift number is right-aligned.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 94         | 95               |
| PERSZEIT  | DEC_O | 10 | 2 | Personal processing time (hours)<br>For postings of the interface record type "A":<br>If this posting record is a U record (interrupt order) or E record (log off order), this field includes the labor time recorded since this record is available. This field includes the logon times of all persons who have logged on to this operation.<br>This field does not include a duration, if it is a T record (reporting part quantities/partial upload), H record (batch posting) or an operation logon.<br>For postings of the interface record type "P":<br>The field includes the length of time a person was logged on to an order. If a person is logged on to several orders/operations simultaneously, the field shows the proportionate labor time per order/operation. The distribution is made proportionally according to the number of operations to which the person is/was logged on at the same time (proportionate labor time). | 96         | 106              |
|           |       |    |   | <b>Resource Performance Accounts (RPA)</b><br>Times are posted to "resource performance accounts". All times refer to the last confirmation/upload performed for the respective operation.<br>The system returns order-related RPAs for postings based on orders (interface record type "A").<br>The system returns personal RPAs for personal postings (record type "P").                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |                  |
| RPA01     | DEC_O | 7  | 2 | Resource Performance Account (RPA) 1 in hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 107        | 114              |
| RPA02     | DEC_O | 7  | 2 | Resource Performance Account (RPA) 2 in hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 115        | 122              |
| RPA03     | DEC_O | 7  | 2 | Resource Performance Account (RPA) 3 in hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 123        | 130              |
| RPA04     | DEC_O | 7  | 2 | Resource Performance Account (RPA) 4 in hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 131        | 138              |
| RPA05     | DEC_O | 7  | 2 | Resource Performance Account (RPA) 5 in hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 139        | 146              |
| RPA06     | DEC_O | 7  | 2 | Resource Performance Account (RPA) 6 in hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 147        | 154              |
| RPA07     | DEC_O | 7  | 2 | Resource Performance Account (RPA) 7 in hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 155        | 162              |
| RPA08     | DEC_O | 7  | 2 | Resource Performance Account (RPA) 8 in hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 163        | 170              |
| RPA09     | DEC_O | 7  | 2 | Resource Performance Account (RPA) 9 in hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 171        | 178              |
| RPA10     | DEC_O | 7  | 2 | Resource Performance Account (RPA) 10 in hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 179        | 186              |

| Field       | T     | L  | D | Description                                                                                                                                                                                                                                                                                                                                                                              | From<br>15 | To <sup>16</sup> |
|-------------|-------|----|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|
| RPA11       | DEC_O | 7  | 2 | Resource Performance Account (RPA) 11 in hours<br><br>RPA 11 contains the production time (standard use/main utilization time).                                                                                                                                                                                                                                                          | 187        | 194              |
| RPA12       | DEC_O | 7  | 2 | Resource Performance Account (RPA) 12 in hours.<br><br>RPA 12 contains neutral times (breaks or similar)                                                                                                                                                                                                                                                                                 | 195        | 202              |
| BMK_SUM     | DEC_O | 7  | 2 | Total of resource performance accounts without breaks (RPA 12) in hours                                                                                                                                                                                                                                                                                                                  | 203        | 210              |
| FIRMA       | CHAR  | 4  |   | reserved                                                                                                                                                                                                                                                                                                                                                                                 | 211        | 214              |
| GRUPPE      | CHAR  | 8  |   | Group where the workplace/machine is assigned to.                                                                                                                                                                                                                                                                                                                                        | 215        | 222              |
| ARBPL       | CHAR  | 8  |   | Workplace/machine this posting refers to.                                                                                                                                                                                                                                                                                                                                                | 223        | 230              |
| PERSNR      | CHAR  | 10 |   | Personnel number.<br>This data field includes the personnel number of the employee who logged on/off or interrupted the operation. In case of personal postings, this field includes the personnel number of the person for whom data is uploaded/confirmed.<br>The precise length depends on the length configuration of the personnel number in the basic parameter settings of HYDRA. | 231        | 240              |
| LOHNART     | CHAR  | 4  |   | Wage type if stored to the operation. Is directly taken over from the operation .                                                                                                                                                                                                                                                                                                        | 241        | 244              |
| GUT_BAS     | DEC_O | 13 | 3 | Basic quantity of yield, if entered or calculated according to conversion factors.                                                                                                                                                                                                                                                                                                       | 245        | 258              |
| AUS_BAS     | DEC_O | 13 | 3 | Basic quantity of scrap, if entered or calculated according to conversion factors.                                                                                                                                                                                                                                                                                                       | 259        | 272              |
| MEINH_BAS   | CHAR  | 3  |   | Basic quantity of the quantity unit, if stored to the operation.                                                                                                                                                                                                                                                                                                                         | 273        | 275              |
| GUT_PRI     | DEC_O | 13 | 3 | Collected yield in primary quantity unit<br>Yield recorded in primary quantity unit since the last upload.                                                                                                                                                                                                                                                                               | 276        | 289              |
| AUS_PRI     | DEC_O | 13 | 3 | Scrap quantity collected in primary quantity unit since the last upload.                                                                                                                                                                                                                                                                                                                 | 290        | 303              |
| MEINH_PRI   | CHAR  | 3  |   | Primary unit of entry (primary quantity unit) from the operation.                                                                                                                                                                                                                                                                                                                        | 304        | 306              |
|             |       |    |   | <b>Reasons</b><br>Reasons are only transferred if the option "confirmation of partial confirmations" is enabled for the order type and only in case of record type "T" postings. Otherwise the fields are empty.                                                                                                                                                                         |            |                  |
| GUTGR       | CHAR  | 4  |   | Yield reason (deviation reason)                                                                                                                                                                                                                                                                                                                                                          | 307        | 310              |
| GUTGR_EXT   | CHAR  | 5  |   | Yield reason (deviation reason) – external reference                                                                                                                                                                                                                                                                                                                                     | 311        | 315              |
| AUSGR       | CHAR  | 4  |   | Scrap reason                                                                                                                                                                                                                                                                                                                                                                             | 316        | 319              |
| AUSGR_EXT   | CHAR  | 5  |   | Scrap reason – external reference                                                                                                                                                                                                                                                                                                                                                        | 320        | 324              |
| ASTATUS     | CHAR  | 1  |   | Order status (control indicator of the order header status)<br>This data field shows the status of an order. When finishing the last recordable operation of an order, this field has the value "E" (end), otherwise "L" (running). The value of this field is not defined for personal confirmations/uploads.                                                                           | 325        | 325              |
| ANMELD_DAT  | DATE  | 8  |   | Date of the terminal posting (login) in the format YYYYMMDD                                                                                                                                                                                                                                                                                                                              | 326        | 333              |
| ANMELD_ZEIT | TIME  | 6  |   | Time of the terminal posting (login) in the format HHMMSS                                                                                                                                                                                                                                                                                                                                | 334        | 339              |

| Field       | T     | L  | D | Description                                                                         | From<br>15 | To <sup>16</sup> |
|-------------|-------|----|---|-------------------------------------------------------------------------------------|------------|------------------|
| ABMELD_DAT  | DATE  | 8  |   | Date of the terminal posting (logoff) in the format YYYYMMDD                        | 340        | 347              |
| ABMELD_ZEIT | TIME  | 6  |   | Time of the terminal posting (logoff) in the format HHMMSS                          | 348        | 353              |
| CHARGE      | CHAR  | 20 |   | Batch number only relevant in connection with ADE-CHV or MPL                        | 354        | 373              |
| BED_POS     | CHAR  | 10 |   | Entered operator position/function according to the HYDRA configuration, if entered | 374        | 383              |
| LPKZ        | CHAR  | 10 |   | Entered wage/premium indicator according to the HYDRA configuration, if entered     | 384        | 393              |
| SOLL_TE     | DEC_O | 7  | 2 | Target te from the log record (only filled if the LLE-BP license is available)      | 394        | 401              |
| SOLL_TR     | DEC_O | 7  | 2 | Target tr from the log record (only filled if the LLE-BP license is available)      | 402        | 409              |
| SOLL_TEB    | DEC_O | 7  | 2 | Target teb from the log record (only filled if the LLE-BP license is available)     | 410        | 417              |
| SOLL_TRB    | DEC_O | 7  | 2 | Target trb from the log record (only filled if the LLE-BP license is available)     | 418        | 425              |

## 8 Inbound Transactions

### Overview

|                        |                                                                        |
|------------------------|------------------------------------------------------------------------|
| HYDRA menu             | System administration → MES Link Enabling (MLE) → Inbound transactions |
| FEDRA menu             | System administration → MES Link Enabling (MLE) → Inbound transactions |
| Transaction code       | intr                                                                   |
| Function authorization | intr<br>intr.reset (reset transactions)                                |

### Purpose

Inbound transactions provide an overview of the data provided from other systems as well as the result of the inbound processing that follows. The application allows for detailed data to be displayed or new posting processes to be triggered.

### Integration

The function allows for data transferred from other systems to be accessed. Other system might be:

- PPS/ERP systems
- Warehouse/material management systems
- Quality management systems

### Field Descriptions

#### Field Descriptions – Inbound Transactions

##### Transaction number

Unique number that is generated while communicating with the external system.

##### Status

The status represents the result of the last processing step. The used status are visualized as follows:

| Description in the application | Color  | Meaning / usage                                                                      |
|--------------------------------|--------|--------------------------------------------------------------------------------------|
| NEW                            | Yellow | The record has been provided initially. Further processing still have to take place. |



| Description in the application | Color  | Meaning / usage                                                                                                                                              |
|--------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TODO                           | Yellow | It has been tried already, to post the record. This attempt has not been successful (for a certain reason) – the record is available to new posting attempt. |
| REACTIVATED                    | Orange | A record posted already has been marked for an additional posting attempt.                                                                                   |
| IN PROCESS                     | Grey   | The record is posted currently.                                                                                                                              |
| UNKNOWN                        | Blue   | There is no valid posting routine for the record.                                                                                                            |
| DONE ERROR                     | Red    | The record could not be posted successfully.                                                                                                                 |
| DONE                           | Green  | The record could be posted successfully.                                                                                                                     |

#### IDoc type

IDoc type of the transaction (whether or not the field is filled out, depends on the communication type).

#### No. of data records

Number of data records included in the transaction

#### No. of edited DR (data records)

Number of successfully processed data records included in the transaction

#### No. of unknown data records

Number of unknown data records included in the transaction

#### No. of erroneous data records

Number of faulty data records included in the transaction (wrong processing)

#### Time of reception

Date and time when the system received the transaction

#### Editing time

Date and time when the transaction was edited in the system

**IDoc number**

IDoc number of transactions (whether or not the field is filled out depends on the communication type).

**Message type**

Message type of the transaction

**Message function**

Message function of the transaction

**SAP sending port**

SAP sending port – only relevant if the communication with SAP is performed via IDoc

**SAP sending partner type**

SAP sending partner type – only relevant if the communication with SAP is performed via IDocs

**SAP sending partner number**

SAP sending partner number – only relevant if the communication with SAP is performed via IDocs

**Receiver port**

Receiver port – only relevant if the communication with SAP is performed via IDocs

**Receiver partner type**

Receiver partner type – only relevant if the communication with SAP is performed via IDocs

**Receiver partner number**

Receiver partner number – only relevant if the communication with SAP is performed via IDocs

**Reference**

Unique database key

**Duration**

Duration of system processing

**Number of attempts**

Number of processing attempts

**Field Descriptions – Log Table****Application**

Application involved in the processing and editing of the transaction

**Log. System**

Logical system

**Role**

Role of the logical system

**Designation**

Designation of the application

**Transaction number**

Transaction number that has been processed

**Status**

Status of processing

**Reference**

Unique database key

**Program**

Technical name of the application

**Program version**

Program version of the application

**Program date**

Program date of the application

**Log file name / log file size**

Name and size of the log file

**Error file name/error file size**

Name and size of the error file

**Data file name/data file size**

Name and size of the data file

**No. of data records**

Number of data records included in the transaction

**No. of edited data records**

Number of successfully processed data records included in the transaction

**No. of unknown data records**

Number of unknown data records included in the transaction

**No. of faulty data records**

Number of faulty data records included in the transaction (wrong processing)

**Text number**

Currently not used

**Posting**

Currently not used

**Created on**

Point in time when the entry was created

**Toolbar****Reset transaction**

The “reset transaction” button allows for a transaction that has already been processed to be processed again. But this is only possible if the transaction has not yet been archived.

**Data segments**

Data segments may be displayed for a transaction. The system tries to display included application data in relation to individual fields within the data record.

**Log**

Displays the log generated by the respective application.

**Error file**

Displays the error log generated by the respective application.

**Data file**

Displays the data file that is optionally generated by the respective application.

## 9 Outbound Transactions

### Overview

|                        |                                                                         |
|------------------------|-------------------------------------------------------------------------|
| HYDRA menu             | System Administration → MES Link Enabling (MLE) → Outbound transactions |
| FEDRA menu             | System Administration → MES Link Enabling (MLE) → Outbound transactions |
| Transaction code       | outtr                                                                   |
| Function authorization | outtr                                                                   |

### Purpose

Outbound transactions provide an overview of the data provided to other systems as well as the result of the outbound processing that follows. The application allows for detailed data to be displayed or new posting processes to be triggered.

### Integration

The function allows for data transferred to other systems to be accessed. Other system might be:

- PPS/ERP systems
- Warehouse/material management systems
- Quality management systems

### Field Descriptions

#### Field Descriptions – Outbound Transactions

##### Transaction number

Unique number that is generated while communicating with the external system.

##### Status

The status represents the result of the last processing step. The used status are visualized as follows:

| Description in the application | Color  | Meaning / usage                                                                      |
|--------------------------------|--------|--------------------------------------------------------------------------------------|
| NEW                            | Yellow | The record has been provided initially. Further processing still have to take place. |
| TODO                           | Yellow | It has been tried already, to                                                        |

| Description in the application | Color  | Meaning / usage                                                                                                                |
|--------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------|
|                                |        | post the record. This attempt has not been successful (for a certain reason) – the record is available to new posting attempt. |
| REACTIVATED                    | Orange | A record posted already has been marked for an additional posting attempt.                                                     |
| IN PROCESS                     | Grey   | The record is posted currently.                                                                                                |
| UNKNOWN                        | Blue   | There is no valid posting routine for the record.                                                                              |
| DONE ERROR                     | Red    | The record could not be posted successfully.                                                                                   |
| DONE                           | Green  | The record could be posted successfully.                                                                                       |

#### IDoc type

IDoc type of the transactions (whether or not the field is filled out, depends on the communication type).

#### No. of data records

Number of data records included in the transaction

#### No. of edited DR (data records)

Number of successfully processed data records included in the transaction

#### No. of unknown data records

Number of unknown data records included in the transaction

#### No. of erroneous data records

Number of faulty data records included in the transaction (wrong processing)

#### Time of reception

Date and time when the system received the transaction

#### Editing time

Date and time when the transaction was edited in the system

**IDoc number**

IDoc number of transactions (whether or not the field is filled out depends on the communication type).

**Message type**

Message type of the transaction

**Message function**

Message function of the transaction

**SAP sending port**

SAP sending port – only relevant if the communication with SAP is performed via IDoc

**SAP sending partner type**

SAP sending partner type – only relevant if the communication with SAP is performed via IDocs

**SAP sending partner number**

SAP sending partner number – only relevant if the communication with SAP is performed via IDocs

**Receiver port**

Receiver port – only relevant if the communication with SAP is performed via IDocs

**Receiver partner type**

Receiver partner type – only relevant if the communication with SAP is performed via IDocs

**Receiver partner number**

Receiver partner number – only relevant if the communication with SAP is performed via IDocs

**Reference**

Unique database key

**Duration**

Duration of system processing

**Number of attempts**

Number of processing attempts

**Field Descriptions – Log Table****Application**

Application involved in the processing and editing of the transaction

**Log. System**

Logical system

**Role**

Role of the logical system

**Designation**

Designation of the application

**Transaction number**

Transaction number that has been processed

**Status**

Status of processing

**Reference**

Unique database key

**Program**

Technical name of the application

**Program version**

Program version of the application

**Program date**

Program date of the application

**Log file name / log file size**

Name and size of the log file

**Error file name/error file size**

Name and size of the error file

**Data file name/data file size**

Name and size of the data file

**No. of data records**

Number of data records included in the transaction

**No. of edited data records**

Number of successfully processed data records included in the transaction

**No. of unknown data records**

Number of unknown data records included in the transaction

**No. of faulty data records**

Number of faulty data records included in the transaction (wrong processing)



**Text number**

Currently not used

**Posting**

Currently not used

**Created on**

Point in time when the entry was created

**Toolbar****Reset transaction**

Use the “reset transaction” button to process a transaction that has already been processed. This is only possible if the transaction has not yet been archived.

During reactivation, the control record of the transaction is set to the status REACTIVATED. The data records included in the transaction are set to status TODO. Once the data records are transferred and are then included in another transaction, there is no connection between the control record of the original transaction and these data records – you cannot show data records for the original control record any more.

**Data segments**

Data segments may be displayed for a transaction. The system tries to display included application data in relation to individual fields within the data record.

**Log**

Displays the log generated by the respective application.

**Error file**

Displays the error log generated by the respective application.

**Data file**

Displays the data file that is optionally generated by the respective application.

## 10 Protecting fields of planned operations

### Purpose

Use the configuration described in this document to prevent specific data fields of a (planned) operation from being overwritten when the operation is transferred once more via the ERP interface.



This function only affects ANR.MODIFY and/or ANR.UPDATE and operations.

Operations are only updated if the status of the order/operation generally allows it. The configuration described below does not apply if the status (see [order status assignment](#)) cannot be changed in general.

### Requirements

You require the relevant function authorization to access [INI configuration](#) and [INI data configuration](#).

### Procedure from service pack 12 onwards (b\_anr.dll version 8.1.1.354)

Create a new entry in the [INI configuration](#):

| Field name  | Value                |
|-------------|----------------------|
| Name        | BAPINOUPDATE         |
| Description | Enter a description. |

For this entry, create an entry including the following values in [INI data configuration](#):

| Field name | Value                                                                                        |
|------------|----------------------------------------------------------------------------------------------|
| Section    | ANR                                                                                          |
| Key        | List the fields (HYDRA BAPI acronyms) that are not overwritten.                              |
| Value      | The value includes a condition. Enter the condition, for example, as follows:<br>ANR.ATYP=AG |
| Active     | Yes                                                                                          |

Use "@" to separate the single fields or conditions in the fields "key" or "value". The fields and conditions are processed one after the other.

You can define the values for "key" and "value" separately. The entries are processed one after the other. The conditions entered in the "value" field correspond to an AND operation.



As of service pack 12 only use the "@" character as separator if you create new entries or change existing ones. You do not have to change existing configurations (prior to service pack 12). In this case, the "|" character is still supported.



You can enter multiple entries for the function BAPINOUPDATE in the [INI data configuration](#), as you define the values for "key" and "value" separately.

## Procedure up to service pack 11

Create a new entry in the [INI configuration](#):

| Field name  | Value                |
|-------------|----------------------|
| Name        | BAPINOUPDATE         |
| Description | Enter a description. |

For this entry, create an entry including the following values in [INI data configuration](#):

| Field name | Value                                                                                                                    |
|------------|--------------------------------------------------------------------------------------------------------------------------|
| Section    | ANR                                                                                                                      |
| Key        | List the fields (HYDRA BAPI acronyms) that are not overwritten.                                                          |
| Value      | Enter the condition that has to be met to make sure fields will not be overwritten. Enter BAPI acronyms including value. |
| Active     | Yes                                                                                                                      |

Use "|" to separate the single fields or conditions in the fields "key" or "value". The fields and conditions are processed one after the other.



Up to service pack 11 only use the "|" character as separator.

You can define the values for "key" and "value" separately. The entries are processed one after the other. The conditions entered in the "value" field correspond to an AND operation.



You can enter multiple entries for the function BAPINOUPDATE in the [INI data configuration](#), as you define the values for "key" and "value" separately.

If you cannot enter the pipe character ("|") using the GUI, you can still enter the values via the database:

- To do so, create a new entry as described above via the [INI configuration](#). Now use the following SQL statement to determine the internal DB counter for the header entry in the INI configuration:

```
select * from hyd_ini
```

- Determine the value of the "VERWEIS" column for the new entry.
- Create the required entries. Use the following SQL statement to assign the database table fields and application fields as described below:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung, aktiv)
values (<reference from previous SQL>, 'ANR', '<fields to be protected>', '<values>',
'<comment>', 'J')
```

Use the "|" (pipe) character to separate the acronyms of the fields you want to protect and the acronyms of the values.



Use a pipe character "|" to complete the list of the fields you want to protect and the list of values.

| Database field      | Values/content                                                             |
|---------------------|----------------------------------------------------------------------------|
| INI_VERWEIS         | The value of the VERWEIS column identified from the HYD_INI table via SQL. |
| SECTION             | Section                                                                    |
| IDENT               | Key                                                                        |
| VALUE               | Value                                                                      |
| BEMERKUNG (comment) | Comment                                                                    |
| AKTIV               | Active                                                                     |

## List of frequently used acronyms

The following table lists the most frequently used acronyms and their meaning. Please contact MPDV Support if the list does not include the acronym you require.

| Acronym      | Meaning                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------|
| ANR.MGRP     | Machine group                                                                                   |
| ANR.MNR      | Workplace/<br>machine                                                                           |
| ANR.OPT:PLAN | Planning indicator:<br>M      Planned for workplace/machine<br>G      Planned for machine group |
| ANR.DATB     | Start date planned (via HLS)                                                                    |
| ANR.ZEIB     | Start time planned (via HLS)                                                                    |
| ANR.DATE     | End date planned (via HLS)                                                                      |
| ANR.ZEIE     | End time planned (via HLS)                                                                      |

## Example: protect the planned workplace

If the operation is planned on a workstation, you have to prevent the ERP interface from cancelling this planning. To do so, enter the below-mentioned data:

| Field name | Value                          |
|------------|--------------------------------|
| Section    | ANR                            |
| Key        | ANR.MGRP@ANR.MNR@ANR.OPT:PLAN@ |
| Value      | ANR.ATYP=AG@ANR.OPT:PLAN=M@    |
| Active     | Yes                            |

SLQ statement:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung,
aktiv) values (<reference from previous SQL>, 'ANR',
'ANR.MGRP@ANR.MNR@ANR.OPT:PLAN@', 'ANR.ATYP=AG@ANR.OPT:PLAN=M@', '<comment>',
'J')
```

## Example: protect the start/end dates of a planned OP

If the operation is planned on a workstation and, as a result, its start time is specified, you have to prevent the ERP interface from cancelling this planning. To do so, enter the below-mentioned data:

| Field name | Value                                |
|------------|--------------------------------------|
| Section    | ANR                                  |
| Key        | ANR.DATB@ANR.ZEIB@ANR.DATE@ANR.ZEIE@ |
| Value      | ANR.ATYP=AG@ANR.OPT:PLAN=M@          |
| Active     | Yes                                  |

SLQ statement:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung,
aktiv) values (<reference from previous SQL>, 'ANR',
'ANR.DATB@ANR.ZEIB@ANR.DATE@ANR.ZEIE@', 'ANR.ATYP=AG@ANR.OPT:PLAN=M@',
'<comment>', 'J')
```

## 11 Modification to the Order of Uploads

### Usage

Subject to the ERP system's way of processing, problems might occur when uploading time tickets to the ERP system. Issue taken: Operations logged on after a shift change are transferred prior to operations interrupted before the change of shifts.

### Requirements

The upload program myerprck.exe/out as of version V8.1.1.99 or higher must be used.

### Procedure

Add the following entry to the [HYDRA INI configuration](#):

| Parameter name | Value                       |
|----------------|-----------------------------|
| INI name       | SAP                         |
| Section        | TIMETICKET_UPLOAD           |
| Key            | ALTERNATIVE_ORDER_BY_CLAUSE |
| Value          | <must remain empty>         |
| Active         | Yes                         |
| Comment        | Changed sorting order       |

### Result

Once activated, uploads will be sorted and reported as follows:

| Sort sequence | Comment | Internal number <sup>17</sup> |
|---------------|---------|-------------------------------|
| Logoff date   |         | 3                             |
| Logoff time   |         | 4                             |

<sup>17</sup> Internal number refers to numbering used in the SQL program logic. The number is not relevant to the end user.

| Sort sequence             | Comment                                                                                                                                                                                                                                                                                                                                                                                                                | Internal number <sup>17</sup> |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Upload type (record type) | <p>The following order applies for uploading record types:</p> <ol style="list-style-type: none"><li>1. operation logons</li><li>2. cancelled operation logoffs</li><li>3. cancelled partial uploads, interruptions to operations, batch records (H records)</li><li>4. Partial uploads</li><li>5. Staff records (B records)</li><li>6. interrupted operations, operation logoffs, batch records (H records)</li></ol> | 1                             |
| Reference                 |                                                                                                                                                                                                                                                                                                                                                                                                                        | 5                             |
| Order number              |                                                                                                                                                                                                                                                                                                                                                                                                                        | 6                             |

## 12 Test Files

### Overview

Attached to this documentation, you will find test files for the interface EIS-ERP. The attachment is only available, if the documentation is in PDF format.

The documentation [Open PDF attachments](#) describes how to call the attached test files.

The following test files are attached to the PDF document:

| File                     | Type                | Comment                                                                             |
|--------------------------|---------------------|-------------------------------------------------------------------------------------|
| HY72ADRCK_TIMETICKET.DAT | Outbound processing | Sample file for the operation-related upload HYDRA --> ERP in HYDRA standard format |
| HY72PPS.DAT              | Inbound processing  | Sample file for the transfer of order data ERP --> HYDRA in HYDRA standard format   |